Original Article

Public Co-Creation Hubs, Cross-Regional Pipelines, and Dynamic Efficiency in Thin Regional Ecosystems

Author information omitted for double-blind review.

Abstract

Local governments increasingly support co-creation hubs, enterprise hubs, and related collaborative spaces as place-based instruments for strengthening regional innovation. Yet the welfare case for such interventions is unclear in thin regional ecosystems, where the local participant pool is limited and repeated local interaction may quickly become redundant. This paper develops a two-region, two-period model in which a public hub raises local matching within a target region, while a complementary pipeline policy creates cross-regional renewal through outside linkages. The analysis yields three main results. First, a hub is socially efficient only if its local coordination gains are large enough to offset fixed cost and the dynamic redundancy generated by repeated local interaction. Second, short-run success and social efficiency are not equivalent: a hub may appear attractive under visible first-period activity metrics while being socially undesirable once dynamic novelty loss is taken into account. Third, the relevant policy object is often not a hub alone but a policy bundle combining local coordination with cross-regional pipeline formation. A hub that fails as a stand-alone intervention may become socially efficient when paired with sufficiently effective outside-linkage policy. Extensions show that participant composition strengthens the case for hub policy, while digital coordination technologies weaken the case for physical hub creation when they substitute for baseline local matching. The model also generates empirically testable predictions on repeated ties, outside collaboration, and absorptive capacity. The results suggest that public hubs in thin regional ecosystems should be evaluated not as stand-alone venues, but as part of a broader regional strategy that combines local buzz with cross-regional renewal.

Keywords: public co-creation hubs; cross-regional pipelines; regional innovation; place-based policy; local buzz; lock-in; thin regional ecosystems

JEL classification: H70, O31, O38, R11, R12, R58

1 Introduction

Publicly supported co-creation hubs, enterprise hubs, and related collaborative spaces have become increasingly visible instruments of regional development policy. Local governments often view such hubs as a way to reduce interaction frictions among firms, researchers, entrepreneurs, and other regional actors by creating

spaces and programs that facilitate meetings, information exchange, and collaboration. This policy logic is especially attractive in thin regional ecosystems, where the local pool of potentially complementary actors is limited and decentralized coordination may otherwise be weak. Yet the welfare case for hub policy is not obvious in such places. A hub may generate visible early activity while still failing to create enough durable novelty to justify its cost, especially if the same local actors repeatedly interact within a narrow regional pool.

This tension sits naturally within the broader literature on agglomeration and regional coordination. A long tradition in urban and regional economics emphasizes that spatial concentration can raise productivity through matching, learning, and related external economies (Duranton and Puga, 2004; Rosenthal and Strange, 2004). Face-to-face interaction has often been treated as a particularly important channel through which localized economic activity generates value (Storper and Venables, 2004). Related work on local buzz and global pipelines similarly suggests that geographically proximate interaction can be productive, especially when local contact is complemented by connections to outside knowledge and opportunities (Bathelt et al., 2004). More recent evidence from co-working environments also indicates that proximate social interaction can indeed create knowledge spillovers and collaboration gains (Roche et al., 2024). Taken together, these contributions provide a strong prima facie rationale for why a public hub might help a region.

At the same time, the same literature also implies that more local interaction is not always better. Proximity can support learning, but it can also generate lock-in when interaction becomes repetitive and the same actors repeatedly draw on the same local knowledge base (Boschma, 2005; Torre and Rallet, 2005). The regional innovation systems literature likewise stresses that local development depends not only on whether interaction occurs, but also on whether regional systems are able to renew themselves through access to external ideas, organizations, and opportunities (Cooke et al., 1997; Asheim et al., 2011; T"odtling and Trippl, 2005). This concern is especially relevant for place-based policy. Local interventions may correct localized coordination problems, but their welfare effects are conditional and context-dependent rather than automatic (Kline and Moretti, 2014a; Neumark and Simpson, 2015). In the case of public hubs, this raises a simple but underexplored question: when is a publicly supported hub socially efficient in a thin regional ecosystem, and when should it instead be viewed as part of a broader regional policy bundle that also creates external renewal?

This paper develops a simple two-region, two-period theory of public co-creation hubs in thin regional ecosystems. Region A is the target thin region in which a public hub may raise the local matching rate among potential collaborators. Region B is an outside pool from which additional collaborators, ideas, and partnerships may arrive. The model's central distinction is between *local coordination* and *cross-regional renewal*. The hub operates on the first margin by increasing the rate at which potentially valuable local interactions in region A are realized. A complementary *pipeline policy* operates on the second margin by increasing the formation of extra-regional links between region A and outside actors in region B.

The model is motivated by a basic dynamic trade-off. In the short run, a hub improves local coordination by lowering interaction frictions and increasing the number of realized local matches. In the longer run, however, that same increase in local interaction intensity may reduce novelty if repeated interaction within a small regional pool becomes redundant. The policy question is therefore not simply whether a hub raises visible local activity. It is whether the hub's short-run coordination gains are large enough to offset fixed cost and the dynamic losses associated with repeated local ties, and whether those losses can be mitigated by policy that expands cross-regional renewal.

The analysis yields three main results. First, a hub is socially efficient as a stand-alone intervention only if its local coordination gain is large enough to offset both its fixed cost and the dynamic redundancy created by repeated local interaction. This produces a simple threshold characterization in terms of ecosystem thickness,

matching improvement, fixed cost, and the rate at which local novelty decays. Second, short-run success and social efficiency are not equivalent. The paper identifies a parameter region in which a hub is supported under a private or local short-run evaluation rule because it generates visible first-period activity, yet is socially undesirable once dynamic redundancy is taken into account. Third, the relevant policy object is often not a hub alone, but a *policy bundle* combining local coordination with cross-regional pipeline formation. Even when a stand-alone hub fails a welfare test, the hub may become socially efficient when paired with sufficiently effective outside-linkage policy.

These results speak directly to current regional policy practice. In many settings, hubs are assessed using visible indicators such as participation counts, meetings, events, occupancy rates, or early project volume. The model suggests that such measures are informative about short-run coordination gains, but incomplete as welfare metrics. What matters for social efficiency is not only whether the hub increases local interaction, but also whether interaction remains novel over time and whether the policy design creates adequate channels for outside renewal. In this sense, the paper argues that a public hub in a thin regional ecosystem should not be evaluated as a stand-alone building or venue, but as part of a broader regional coordination architecture.

The paper contributes to three strands of literature. Relative to the agglomeration and interaction literatures, it provides a welfare framework for a policy instrument that is often justified by local buzz and face-to-face contact, but rarely analyzed through the joint lens of coordination gains and dynamic redundancy (Duranton and Puga, 2004; Storper and Venables, 2004; Bathelt et al., 2004). Relative to the proximity and regional innovation systems literatures, it connects lock-in and system renewal to an explicit welfare criterion that distinguishes local coordination from cross-regional renewal (Boschma, 2005; Asheim et al., 2011). Relative to the place-based policy literature, it shows why the efficient object of evaluation may be a bundled regional policy rather than a stand-alone local project (Kline and Moretti, 2014a; Neumark and Simpson, 2015).

The model also generates empirically testable implications. It predicts that hub creation should increase initial interaction volume, but not necessarily sustain the share of new ties; that hub-plus-pipeline regions should exhibit stronger outside collaboration and higher medium-run novelty than hub-only regions; and that the same hub policy should be more valuable in regions with greater absorptive capacity. These predictions point toward a more dynamic and network-aware empirical evaluation of hub policy, one that tracks repeated ties, outside linkages, and participant composition rather than relying only on visible activity measures.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature and situates the paper's contribution. Section 3 presents the two-region model. Section 4 characterizes the conditions under which a hub is socially efficient as a stand-alone policy. Section 5 explains why a place-based public intervention may be needed by distinguishing private short-run evaluation from social welfare evaluation. Section 6 analyzes the optimal pipeline policy and the conditions under which a hub-plus-pipeline bundle is efficient. Section 7 develops extensions on participant composition and digital substitution. Section 8 derives empirical predictions and discusses regional policy interpretation. Section 9 concludes.

2 Related Literature and Contribution

This paper relates to three strands of research: the literature on local interaction, agglomeration, and regional coordination; the literature on proximity, lock-in, and system renewal; and the welfare analysis of place-based policy with particular attention to external linkages and regional policy design. The paper's contribution is to connect these strands in a two-region dynamic welfare framework in which a public hub raises local coordination within a target thin region, while a complementary pipeline policy creates cross-regional renewal

through outside linkages.

2.1 Local Interaction, Agglomeration, and Regional Coordination

A first relevant literature explains why spatial concentration can raise productivity. In urban and regional economics, agglomeration economies are commonly rationalized through mechanisms such as matching, learning, and sharing (Duranton and Puga, 2004; Rosenthal and Strange, 2004). In this view, spatial proximity improves the efficiency with which actors find collaborators, exchange information, and combine complementary ideas or capabilities. This logic provides the most direct theoretical motivation for a public co-creation hub: a hub can be understood as a policy device that raises the effective rate at which potentially valuable local interactions are realized within a region.

Within this broader perspective, face-to-face interaction has been emphasized as a distinct source of value. Storper and Venables (2004) argue that face-to-face contact facilitates trust formation, complex communication, and the transfer of tacit knowledge. Jaffe et al. (1993) show that knowledge spillovers are geographically localized, while Roche et al. (2024) provide more direct evidence that proximate co-working environments can generate knowledge spillovers through local social interaction. These contributions support the paper's premise that reducing local interaction frictions may create social value even in relatively small regional ecosystems.

The paper also speaks to the emerging literature on co-working spaces, enterprise hubs, and collaborative spaces outside major metropolitan cores. Vogl and Akhavan (2022) review the evidence on co-working spaces in peripheral and rural areas and conclude that their effects are heterogeneous and context-dependent. Merrell et al. (2022) analyze enterprise hubs as a mechanism for rural local economic development, while Merrell et al. (2021) interpret rural enterprise hubs as micro-clusters that may support creativity-led development. This literature is close to the policy object studied here, but it remains largely empirical and descriptive. It documents possible benefits of hubs and the conditions under which they succeed, but it does not offer a welfare criterion for determining when a publicly supported hub is efficient, when it merely generates visible short-run activity, or when it should be complemented by external-linkage policy.

Relative to these literatures, the present paper does not simply restate that local interaction is valuable. Instead, it embeds a hub in a regional welfare framework in which local coordination is beneficial but not automatically sufficient for dynamic efficiency. In that sense, the paper takes the interaction-based logic of agglomeration seriously, but shifts the question from whether local interaction matters to when a public device that organizes local interaction is socially justified.

2.2 Proximity, Lock-In, and System Renewal

A second relevant literature emphasizes that proximity has both benefits and costs. Boschma (2005) argues that proximity can support innovation, but that too much proximity can generate lock-in by narrowing the scope for new combinations. Torre and Rallet (2005) similarly distinguish different dimensions of proximity and stress that spatial closeness is neither universally beneficial nor sufficient for sustained innovation. This literature is central for the present paper because the model's dynamic redundancy mechanism is precisely a formalization of the idea that repeated local interaction may become less novel over time.

The paper is also closely related to the regional innovation systems literature. Cooke et al. (1997) and Asheim et al. (2011) stress that regional innovation depends on the institutional and organizational structure of

local systems rather than on scale alone. T"odtling and Trippl (2005) argue that innovation policy should be differentiated across places, while Coenen et al. (2017) emphasize evolutionary and system-level perspectives in regional policy. Work on path renewal and path creation further shows that even regions with established specializations may struggle to generate new trajectories without appropriate renewal mechanisms (Chaminade et al., 2019).

What matters for the present paper is that this literature does more than warn against excessive proximity. It also points toward the importance of renewal mechanisms that reconnect regions to outside sources of knowledge, collaborators, and opportunities. Bathelt et al. (2004) capture this especially well through the distinction between local buzz and global pipelines. That distinction is highly relevant here. In the model developed below, the public hub plays the role of a local coordination device inside the target region, while the pipeline policy captures a distinct margin of adjustment: the formation of extra-regional links that replenish novelty and reduce the danger of local lock-in.

This is the key conceptual bridge between the regional innovation systems literature and the present model. The model does not treat renewal as a vague narrative possibility. Rather, it treats renewal as a policy-relevant margin that can be compared directly with local coordination in welfare terms. Put differently, the paper links the literature on proximity and lock-in to an explicit criterion for deciding when local interaction should be supplemented by cross-regional renewal.

2.3 Place-Based Policy, External Linkages, and This Paper's Contribution

A third relevant literature studies the welfare properties of geographically targeted policy. Kline and Moretti (2014a) provide a general welfare-theoretic treatment of local economic development programs, emphasizing that place-based interventions may correct localized distortions but need not be efficient in general. Kline and Moretti (2014b) show in a major historical application that geographically targeted interventions can produce long-run local gains, while also making clear that the welfare case for such interventions depends on the underlying externalities and margins of adjustment. More broadly, Neumark and Simpson (2015) survey place-based policies and stress that their effects are heterogeneous and context-dependent.

The present paper belongs to this place-based policy tradition, but studies a policy instrument that differs from the tax incentives, subsidies, and infrastructure programs more commonly analyzed in that literature in that the object here is a public hub whose purpose is to reduce local interaction frictions in a thin target region. More importantly, the paper argues that the relevant policy object is often not the hub alone, but a *policy bundle* that combines local coordination with cross-regional linkage formation. This is where the two-region structure matters. Region A is the target thin region in which the hub raises local matching; region B is an outside pool from which additional collaborations, ideas, and partnerships may arrive. The pipeline policy is therefore not an abstract add-on, but a reduced-form representation of extra-regional linkage formation, external brokerage, and system-renewal mechanisms.

This perspective also sharpens the rationale for public intervention. A short-run private or local evaluation rule may focus on visible activity generated by the hub inside region A, such as meetings, participation, or local matches. A social planner, by contrast, evaluates the hub more broadly, incorporating both the dynamic consequences of repeated local ties and the cross-regional spillovers generated by outside linkages. The paper's contribution is to formalize this divergence without requiring a full decentralized equilibrium model. In the model, a hub may look attractive under a short-run local criterion yet be socially undesirable once dynamic redundancy is taken into account. Conversely, a hub may fail a short-run private test and still be

socially desirable once paired with cross-regional pipeline policy.

Against this background, the paper makes three contributions. First, it introduces an explicitly regional distinction between *local coordination* and *cross-regional renewal*. The hub raises local matching within the target region, whereas the pipeline policy creates or intensifies extra-regional links with an outside pool. Second, it provides a welfare criterion for evaluating hub policy as a *policy bundle*, thereby connecting local buzz, lock-in, and external renewal within a single framework. Third, it generates empirically testable predictions on repeated ties, outside linkages, and participant composition. In that sense, the paper contributes not only to the theory of place-based policy, but also to the empirical agenda on how hubs and related coordination devices should be evaluated in thin regional ecosystems.

The paper therefore contributes to the agglomeration literature by formalizing a public interaction device, to the proximity and regional innovation systems literatures by connecting lock-in and renewal to explicit welfare comparisons, and to the place-based policy literature by showing why the efficient object of evaluation may be a bundled regional policy rather than a stand-alone local project.

3 The Model

We consider a target thin region A and an outside region B. A local government in region A decides whether to establish a public co-creation hub and, if a hub is in place, whether to supplement it with a cross-regional pipeline policy. The model preserves the paper's two-period structure, but it now distinguishes explicitly between *local coordination within the target region* and *cross-regional renewal from outside the region*. This is the minimal extension needed to evaluate hub policy as a regional policy bundle rather than as a stand-alone local intervention.

3.1 Environment: Two Regions and Potential Interactions

Region A contains $N_A \geq 2$ local participants, where a participant may be interpreted as a firm, researcher, entrepreneur, or other actor capable of generating collaborative surplus. Region B contains $N_B \geq 1$ outside participants who form the external pool from which new ideas, collaborators, and partnerships may arrive.

The number of unordered potential *local* pairs within region A is

$$P_A := \frac{N_A(N_A - 1)}{2}, \tag{1}$$

and the number of potential *cross-regional* links between regions A and B is

$$Q_{AB} := N_A N_B. \tag{2}$$

A benevolent local planner in region A chooses whether to establish a public hub. Let

$$h \in \{0, 1\}$$

denote the hub decision, where $h = 1$ means that a public hub is established and $h = 0$ means that it is not.

If a hub exists, the planner may also implement a cross-regional pipeline policy,

$$b \geq 0,$$

which captures efforts to connect region A more intensively to outside participants in region B. Examples include guest programs, interregional brokerage, visiting mentors, university–industry linkage schemes, and outreach to firms or researchers located outside the target region.

Time is discrete and consists of two periods, $t = 1, 2$. The hub affects local matching in both periods. The pipeline policy affects only period 2.

3.2 Local Coordination Through a Hub

Let $m_A(h)$ denote the effective local matching rate among potential pairs within region A. We assume

$$m_A(h) = m_A^0 + \left(m_A^1 - m_A^0\right)h, \qquad 0 \leq m_A^0 < m_A^1 \leq 1. \tag{3}$$

Thus, establishing a hub lowers local interaction costs in region A and raises the fraction of potential local pairs that become realized economically meaningful interactions.

Let $v_L > 0$ denote the average social surplus generated by one realized and still-novel *local* interaction. Then period-1 gross surplus is

$$B_1(h) = v_L P_A m_A(h). \tag{4}$$

Equation (4) captures the hub's immediate coordination role inside the target region: by lowering local search and meeting frictions, it increases the number of productive local encounters in the short run.

3.3 Dynamic Redundancy in Local Interactions

The same local interactions that create value in period 1 may reduce novelty in period 2 if the local network becomes too repetitive. To capture this with minimal structure, let the period-2 *local* component of surplus be

$$B_{2L}(h) = \beta v_L P_A m_A(h)\left[1 - \delta m_A(h)\right], \qquad \beta \in (0, 1],\ \delta > 0. \tag{5}$$

The parameter δ measures the rate at which intensive local interaction generates redundancy, repeated exposure, or local lock-in. A higher δ means that a denser round of local interaction in region A reduces second-period novelty more sharply.

Expression (5) may be read as a reduced-form summary of repeated local matching. When the hub raises $m_A(h)$, it increases both the number of realized local interactions and the probability that some of those interactions become less novel in period 2 because they draw repeatedly on the same local pool.

We maintain the regularity condition

$$1 - \delta m_A^1 > 0, \tag{6}$$

which ensures that second-period local novelty remains positive even under the hub.

3.4 Cross-Regional Pipeline Policy

In period 2, the planner may also create additional links between region A and outside participants in region B. Let the effective cross-regional connection rate (or intensity) be

$$q(b) = q_0 + \eta bh, \qquad q_0 \geq 0, \quad \eta > 0. \tag{7}$$

The baseline term q_0 captures cross-regional access that exists even without new policy. The term ηbh captures the idea that the pipeline policy raises region A's outside connectivity only when a hub is present to organize, absorb, and channel those external contacts. Accordingly, if $h = 0$, then $q(b) = q_0$ for all b, so it is without loss to set $b = 0$.

Let $v_X > 0$ denote the average social surplus generated by one effective *cross-regional* linkage between regions A and B. The period-2 *external* component of surplus is then

$$B_{2X}(h, b) = \beta v_X Q_{AB} q(b). \tag{8}$$

Combining (5) and (8), total period-2 gross surplus is

$$B_2(h, b) = \beta\Big\{v_L P_A m_A(h)\big[1 - \delta m_A(h)\big] + v_X Q_{AB} q(b)\Big\}. \tag{9}$$

Substituting (7) into (9) gives

$$B_2(h, b) = \beta\Big\{v_L P_A m_A(h)\big[1 - \delta m_A(h)\big] + v_X Q_{AB}\big(q_0 + \eta bh\big)\Big\}. \tag{10}$$

This formulation is the key change relative to the single-region baseline. The hub continues to operate through local coordination in region A, while the pipeline policy now has a concrete regional interpretation: it raises the rate of cross-regional connections between the target region and an outside pool.

3.5 Welfare, Timing, and Useful Benchmark Expressions

Establishing the hub requires a fixed cost $F > 0$. The pipeline policy has convex cost

$$\frac{\kappa}{2} b^2, \qquad \kappa > 0.$$

Because the pipeline policy is relevant only when a hub exists, it is convenient to write its cost as $\frac{\kappa}{2} b^2 h$.

Total social welfare is therefore

$$W(h, b) = B_1(h) + B_2(h, b) - Fh - \frac{\kappa}{2} b^2 h. \tag{11}$$

Using (4) and (10), this can be written as

$$W(h, b) = v_L P_A m_A(h) + \beta\Big\{v_L P_A m_A(h)\big[1 - \delta m_A(h)\big] + v_X Q_{AB}\big(q_0 + \eta bh\big)\Big\} - Fh - \frac{\kappa}{2} b^2 h. \tag{12}$$

The planner's problem is to choose (h, b) to maximize (12), taking $(N_A, N_B, m_A^0, m_A^1, v_L, v_X, \delta, q_0, \eta, F, \kappa)$ as given.

The timing is as follows.

1. At the beginning of period 1, the planner chooses whether to establish a public hub, $h \in \{0, 1\}$.
2. Period-1 local matching occurs in region A, and period-1 surplus $B_1(h)$ is realized.
3. At the beginning of period 2, if $h = 1$, the planner chooses the cross-regional pipeline policy $b \geq 0$.
4. Period-2 local and cross-regional surplus, summarized by $B_2(h, b)$, is realized.

Since there is no uncertainty, this staged timing is equivalent to choosing (h, b) at the outset under commitment. The sequential formulation is adopted only because it highlights the interpretation of b as a policy response to the hub's dynamic tendency toward local redundancy.

For later use, it is convenient to record the welfare levels associated with the three benchmark cases.

First, if no hub is established, then $h = 0$ and $b = 0$, so

$$W(0,0) = v_L P_A m_A^0 + \beta\Big\{v_L P_A m_A^0 \big(1 - \delta m_A^0\big) + v_X Q_{AB} q_0\Big\}. \tag{13}$$

Second, if a hub is established but no pipeline policy is used, then

$$W(1,0) = v_L P_A m_A^1 + \beta\Big\{v_L P_A m_A^1 \big(1 - \delta m_A^1\big) + v_X Q_{AB} q_0\Big\} - F. \tag{14}$$

Third, if a hub is established and pipeline policy b is used, then

$$W(1,b) = W(1,0) + \beta\eta v_X Q_{AB} b - \frac{\kappa}{2} b^2. \tag{15}$$

Subtracting (13) from (14), the net welfare gain from a *hub alone* is

$$\Delta_H := W(1,0) - W(0,0) = v_L P_A \big(m_A^1 - m_A^0\big)\Big[(1+\beta) - \beta\delta\big(m_A^1 + m_A^0\big)\Big] - F. \tag{16}$$

Subtracting (13) from (15), the net welfare gain from a *hub with pipeline policy* is

$$\Delta_{H+b}(b) := W(1,b) - W(0,0) = \Delta_H + \beta\eta v_X Q_{AB} b - \frac{\kappa}{2} b^2. \tag{17}$$

Equations (16) and (17) already show the model's basic logic. The stand-alone value of the hub is governed by *local* coordination inside region A and the associated risk of dynamic redundancy. By contrast, the incremental value of the pipeline policy is governed by the strength of *cross-regional* renewal through additional A–B linkages. The baseline outside connection rate q_0 affects levels of welfare, but it cancels out of the incremental hub-alone comparison.

The next section uses (16) to characterize when a public hub is socially efficient as a stand-alone policy. Section 5 then distinguishes short-run success from dynamic efficiency. Section 6 introduces optimal pipeline policy and characterizes when a hub alone is undesirable but a hub combined with cross-regional renewal is socially efficient.

4 When Is a Public Hub Efficient?

This section characterizes the conditions under which a public hub is socially efficient *as a stand-alone local policy*. Thus, throughout this section, we set

$$b = 0.$$

The purpose is to isolate the hub's value through local coordination inside region A before turning in Section 6 to the case in which the hub is combined with a cross-regional pipeline policy.

4.1 A Benchmark Efficiency Criterion

From (16), the net welfare gain from establishing a hub without pipeline policy is

$$\Delta_H = v_L P_A \left(m_A^1 - m_A^0\right)\left[(1+\beta) - \beta\delta\left(m_A^1 + m_A^0\right)\right] - F. \tag{18}$$

It is useful to define the hub's *dynamic viability factor* as

$$\Lambda_A := (1+\beta) - \beta\delta\left(m_A^1 + m_A^0\right). \tag{19}$$

Then (18) becomes

$$\Delta_H = v_L P_A \left(m_A^1 - m_A^0\right)\Lambda_A - F. \tag{20}$$

The term $m_A^1 - m_A^0$ captures the hub's immediate local coordination value: how much it raises the effective matching rate within region A. By contrast, Λ_A captures whether those short-run gains remain socially valuable once second-period local redundancy is taken into account. A larger δ lowers Λ_A, making the hub less attractive from a dynamic welfare perspective.

The hub is socially efficient if and only if

$$\Delta_H \geq 0.$$

Proposition 1. *Consider the benchmark case without pipeline policy, $b = 0$.*

1. *If*

$$\Lambda_A \leq 0,$$

then the hub is never socially efficient:

$$\Delta_H < 0 \qquad \textit{for all } N_A \geq 2 \textit{ and } F > 0.$$

2. *If*

$$\Lambda_A > 0,$$

then the hub is socially efficient if and only if

$$P_A \geq \frac{F}{v_L\left(m_A^1 - m_A^0\right)\Lambda_A}. \tag{21}$$

Equivalently, treating N_A as continuous, the hub is efficient if and only if

$$N_A \geq N_H^* := \frac{1}{2}\left(1 + \sqrt{1 + \frac{8F}{v_L(m_A^1 - m_A^0)\Lambda_A}}\right). \tag{22}$$

If N_A is restricted to integers, replace N_H^* by its ceiling.

Proposition 1 has a simple interpretation. A hub creates value only if the gross local coordination benefit,

$$v_L P_A(m_A^1 - m_A^0),$$

survives the dynamic penalty from repeated local interaction, summarized by Λ_A, and is large enough to cover the fixed cost F. Thus, a hub may fail to be efficient for two conceptually distinct reasons. First, the target region may be too thin, so that P_A is too small. Second, even if the region is not especially thin, the redundancy parameter δ may be large enough to make repeated local interaction dynamically unproductive.

A useful implication of the two-region formulation is that the stand-alone criterion in (20) depends only on *local* primitives. The size of the outside pool N_B, the baseline outside access q_0, and the surplus attached to cross-regional links v_X affect welfare levels in (13) and (14), but they cancel out of the incremental comparison between a hub alone and no hub.

4.2 Equivalent Threshold Characterizations

Proposition 1 can be rewritten in several equivalent ways, which are useful for interpretation.

First, for given $(N_A, m_A^0, m_A^1, \beta, \delta)$, define the critical fixed cost

$$F_H^*(N_A) := v_L P_A(m_A^1 - m_A^0)\Lambda_A. \tag{23}$$

Then the hub is efficient if and only if

$$F \leq F_H^*(N_A).$$

Second, for given $(N_A, m_A^0, m_A^1, \beta, F)$, define the critical redundancy rate

$$\delta_H^*(N_A) := \frac{1+\beta}{\beta(m_A^1 + m_A^0)} - \frac{F}{\beta v_L P_A(m_A^1 - m_A^0)(m_A^1 + m_A^0)}. \tag{24}$$

Then the hub is efficient if and only if

$$\delta \leq \delta_H^*(N_A),$$

provided the right-hand side is positive.

These equivalent thresholds make clear that the hub's stand-alone efficiency is governed by five local primitives: the size of the target ecosystem N_A, the hub's matching improvement $m_A^1 - m_A^0$, the surplus per realized local interaction v_L, the fixed cost F, and the speed of local redundancy δ.

4.3 Comparative Statics

The next proposition summarizes how the hub's net welfare contribution responds to the primitive parameters.

Proposition 2. *Suppose $\Lambda_A > 0$. Then the hub-alone welfare gain Δ_H satisfies*

$$\frac{\partial \Delta_H}{\partial N_A} = \frac{v_L}{2}(2N_A - 1)\left(m_A^1 - m_A^0\right)\Lambda_A > 0, \tag{25}$$

$$\frac{\partial \Delta_H}{\partial F} = -1 < 0, \tag{26}$$

$$\frac{\partial \Delta_H}{\partial \delta} = -\beta v_L P_A\left(m_A^1 - m_A^0\right)\left(m_A^1 + m_A^0\right) < 0. \tag{27}$$

Moreover, under the regularity condition (6)*,*

$$\frac{\partial \Delta_H}{\partial m_A^1} = v_L P_A\left[(1+\beta) - 2\beta\delta m_A^1\right] > 0, \tag{28}$$

$$\frac{\partial \Delta_H}{\partial m_A^0} = -v_L P_A\left[(1+\beta) - 2\beta\delta m_A^0\right] < 0. \tag{29}$$

Proposition 2 formalizes several intuitive points.

First, the hub is more likely to be efficient in a thicker target ecosystem. Since

$$P_A = \frac{N_A(N_A - 1)}{2},$$

the number of potential local collaborative pairs rises more than proportionally with N_A. This is the sense in which thin regional ecosystems face a structural disadvantage: even if the hub improves local matching, there may simply be too few potentially valuable local interactions for the gain to cover the fixed cost.

Second, faster local redundancy makes the hub less attractive. This is the model's formal representation of local lock-in or repeated interaction among the same actors inside region A. A hub is therefore not justified by short-run coordination gains alone; its dynamic viability depends on whether those gains persist.

Third, a stronger local matching effect of the hub, measured by a larger m_A^1, raises the hub's welfare contribution, whereas a higher baseline local matching rate m_A^0 lowers its incremental value. Thus, a public hub is most useful where pre-existing local coordination is weak but where the hub can substantially lower interaction frictions.

4.4 Discussion

This section yields two implications that will be important later.

The first is that hub efficiency is not determined by local scale alone. Even a relatively large target region may not justify a hub if local redundancy builds quickly. Conversely, even a thin region may justify a hub if the coordination gain is large enough and dynamic redundancy is limited.

The second is that hub efficiency depends on a dynamic object, Λ_A, rather than only on the static matching improvement $m_A^1 - m_A^0$. This foreshadows the central message of the paper: a public hub can look attractive on short-run local coordination grounds and yet be dynamically inefficient once the risk of repeated interaction within the same regional pool is taken seriously.

For that reason, the next section distinguishes explicitly between short-run success and total dynamic

efficiency.

5 Why Public Intervention? Private Evaluation and Social Welfare

Sections 4 and 6 study welfare under a benevolent planner. For the paper's policy interpretation, however, it is also important to explain why a place-based public intervention may be needed in the first place. In practice, public hubs are often advocated or opposed using highly visible short-run indicators such as participation counts, meetings, first-round matches, or occupancy rates. These criteria do not necessarily coincide with social welfare.

This section therefore introduces a minimal contrast between a *private evaluation rule* and a *social evaluation rule*. The point is not to build a full decentralized equilibrium model. Rather, it is to show as parsimoniously as possible that local private stakeholders may evaluate a hub through visible short-run activity, whereas a social planner also values dynamic novelty and cross-regional spillovers. The two criteria need not coincide.

5.1 A Private Evaluation Rule Based on Visible Short-Run Activity

Suppose that a private hub operator, local coalition, or other decentralized stakeholder evaluates the hub primarily through visible short-run activity in region A. Under the present two-period setup, the natural reduced-form counterpart is the hub's period-1 net gain:

$$\Pi_H := \left[B_1(1) - F\right] - B_1(0). \tag{30}$$

Using (4), this becomes

$$\Pi_H = v_L P_A\left(m_A^1 - m_A^0\right) - F. \tag{31}$$

Expression (31) is the model's reduced-form representation of a private short-run evaluation rule. It depends only on the hub's immediate contribution to local matching in region A, net of the fixed cost. It ignores both dynamic redundancy in repeated local interaction and any social value created by future cross-regional linkages.

A hub is therefore *privately supported under short-run evaluation* if and only if

$$\Pi_H > 0.$$

Proposition 3. *Under the short-run private evaluation rule* (30)*, a public hub is privately supported if and only if*

$$F < v_L P_A\left(m_A^1 - m_A^0\right). \tag{32}$$

Equivalently, treating N_A as continuous, private support obtains if and only if

$$N_A \geq N_{SR}^* := \frac{1}{2}\left(1 + \sqrt{1 + \frac{8F}{v_L\left(m_A^1 - m_A^0\right)}}\right), \tag{33}$$

with the usual ceiling operator if N_A is restricted to integers.

Proposition 3 depends only on the local period-1 coordination gain. In that sense, it is deliberately myopic.

This is useful because many real-world arguments for or against hubs are framed in exactly these terms.

5.2 A Social Evaluation Rule

The planner's criterion is broader. As shown in Sections 4 and 6, social welfare depends not only on immediate local coordination, but also on dynamic redundancy in local interactions and on cross-regional renewal.

For notational convenience, define

$$S_A := v_L P_A\big(m_A^1 - m_A^0\big) > 0, \tag{34}$$

which is the hub's visible short-run local coordination gain,

$$D_A := \beta v_L P_A\big(m_A^1 - m_A^0\big)\Big[1 - \delta\big(m_A^1 + m_A^0\big)\Big], \tag{35}$$

which is the dynamic correction generated by second-period local interaction, and

$$S_{AB}(b) := \beta\eta v_X Q_{AB}\, b - \frac{\kappa}{2} b^2, \tag{36}$$

which is the incremental cross-regional renewal surplus generated by pipeline policy b.

Using (18) and (46), the hub-alone and hub-plus-pipeline welfare gains can be written as

$$\Delta_H = \Pi_H + D_A, \tag{37}$$

and

$$\Delta_{H+b}(b) = \Pi_H + D_A + S_{AB}(b). \tag{38}$$

Equation (37) shows that the social value of a hub alone equals the private short-run criterion plus a dynamic correction term. Equation (38) adds a third component: cross-regional renewal. The normative justification for public intervention is exactly that decentralized short-run evaluation omits these additional social terms.

Moreover, under the private short-run rule (30), pipeline policy has no direct private value because it affects neither period-1 local activity nor the private criterion itself. Hence a purely short-run private evaluator would set

$$b^P = 0,$$

or at least would have no reason to choose a positive b. By contrast, the planner values b through (36) and chooses the positive optimum characterized in Section 6.

Let

$$R_{AB}(N_A, N_B) := \frac{\big(\beta\eta v_X Q_{AB}\big)^2}{2\kappa} = \frac{\beta^2\eta^2 v_X^2 N_A^2 N_B^2}{2\kappa} > 0 \tag{39}$$

denote the maximized cross-regional renewal surplus from Section 6. Then

$$\Delta_{H+b}^* = \Pi_H + D_A + R_{AB}(N_A, N_B). \tag{40}$$

5.3 When Do Private and Social Evaluations Diverge?

The following proposition collects the two most important cases.

Proposition 4. *Let*

$$F_H^*(N_A) = S_A + D_A \qquad \textit{and} \qquad F_{H+b}^*(N_A, N_B) = S_A + D_A + R_{AB}(N_A, N_B).$$

1. ***Private over-support of a hub alone.*** *If*

$$\delta\big(m_A^1 + m_A^0\big) > 1, \tag{41}$$

then $D_A < 0$, so

$$\Delta_H < \Pi_H.$$

Moreover, there exists a nonempty interval of fixed costs for which the hub is privately supported under short-run evaluation but socially undesirable as a stand-alone policy:

$$\max\big\{0,\ F_H^*(N_A)\big\} < F < S_A. \tag{42}$$

2. ***Private under-support of a hub-plus-pipeline bundle.*** *If*

$$D_A + R_{AB}(N_A, N_B) > 0, \tag{43}$$

equivalently if

$$F_{H+b}^*(N_A, N_B) > S_A,$$

then there exists a nonempty interval of fixed costs for which the hub is not privately supported under short-run evaluation, yet a hub combined with optimally chosen pipeline policy is socially desirable:

$$S_A \le F < F_{H+b}^*(N_A, N_B). \tag{44}$$

Proposition 4 gives a compact rationale for place-based public intervention.

Part (1) shows that a hub may be *privately attractive* because it generates visible local activity, yet *socially undesirable* because repeated local interaction destroys too much future novelty. In this case, a purely decentralized short-run criterion overstates the hub's value.

Part (2) shows the opposite possibility. Once dynamic and cross-regional gains are taken into account, a hub combined with pipeline policy may be *socially desirable* even though it does not pass the private short-run test. In this case, decentralized evaluation understates the value of the broader regional policy bundle.

Together, these two cases imply that the relevant policy question is not simply whether local actors can justify a hub using immediate activity metrics. The deeper question is whether the hub, viewed as part of a place-based coordination architecture, creates enough long-run and cross-regional value to justify public intervention.

5.4 Interpretation

The section delivers three implications.

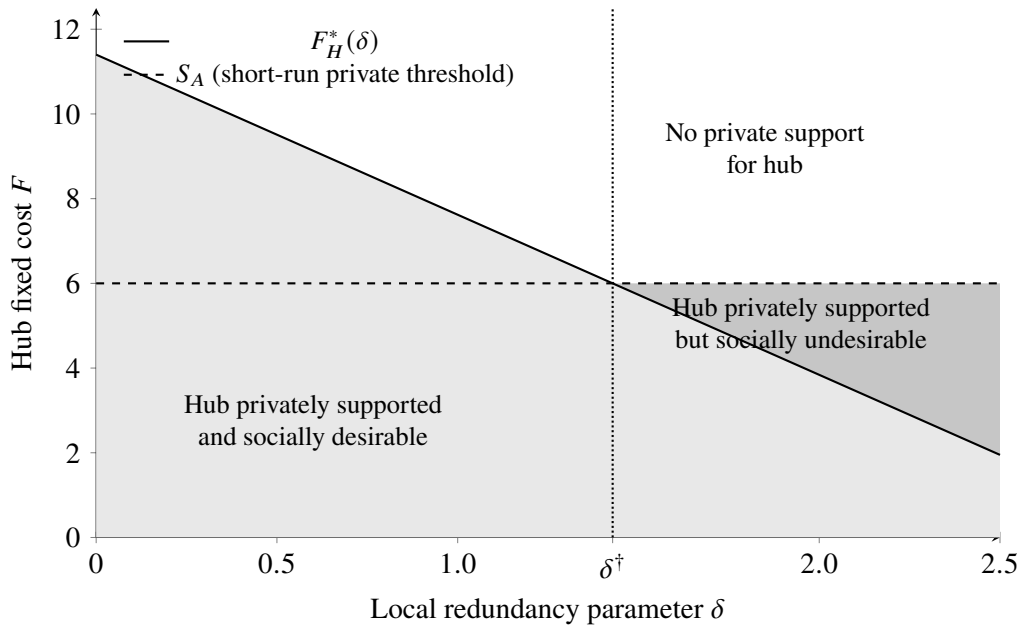

Figure 1: Illustrative policy regions in (δ, F) space. The solid line shows the social efficiency threshold for a hub alone, $F_H^*(\delta)$, while the dashed line shows the short-run private support threshold, $S_A = v_L P_A(m_A^1 - m_A^0)$. For $\delta > \delta^\dagger = 1/(m_A^1 + m_A^0)$, there exists a region in which the hub is privately supported under short-run evaluation but socially undesirable once dynamic redundancy is taken into account. Parameter values are illustrative: $N_A = 6$, $v_L = 1$, $m_A^0 = 0.15$, $m_A^1 = 0.55$, and $\beta = 0.9$.

First, a short-run private rule naturally ignores dynamic redundancy. This creates the possibility that local stakeholders favor a hub because it produces visible activity today, even though it lowers social welfare once repeated local matching is taken into account.

Second, a short-run private rule also ignores cross-regional spillovers. This creates the possibility that local stakeholders reject a hub because its visible short-run returns are too small, even though a hub combined with outside-linkage policy would be socially beneficial.

Third, the role of government in the present framework is therefore not merely to finance an otherwise identical private project. It is to evaluate the hub on a wider margin than decentralized actors do: not only by immediate local coordination, but also by dynamic renewal and by the creation of cross-regional pipelines. This is the paper's minimal theoretical rationale for public intervention.

Section 6 builds directly on this logic by characterizing the optimal cross-regional pipeline policy and the conditions under which the efficient policy object is not a hub alone, but a hub combined with external linkage formation.

6 Hub Alone vs. Hub Plus Cross-Regional Pipeline Policy

Sections 4 and 5 showed that a public hub may fail on dynamic grounds even when it looks attractive in the short run. This section asks whether that weakness can be corrected by a *cross-regional pipeline policy* that brings in outside participants, ideas, or partnerships in period 2.

We now restore the policy variable $b \geq 0$. Throughout, we compare three objects:

$$W(0,0), \qquad W(1,0), \qquad W(1,b).$$

The first is the no-hub benchmark, the second is a hub alone, and the third is a hub combined with cross-regional pipeline policy.

6.1 Optimal Pipeline Policy Conditional on a Hub

From (15), welfare with a hub and pipeline policy can be written as

$$W(1,b) = W(1,0) + \beta\eta v_X Q_{AB}\, b - \frac{\kappa}{2} b^2. \tag{45}$$

Equivalently, using (17),

$$\Delta_{H+b}(b) := W(1,b) - W(0,0) = \Delta_H + \beta\eta v_X Q_{AB}\, b - \frac{\kappa}{2} b^2. \tag{46}$$

Thus, conditional on $h = 1$, the planner solves a strictly concave quadratic problem in b.

Proposition 5. *Conditional on establishing a hub, the unique welfare-maximizing pipeline policy is*

$$b^* = \frac{\beta\eta v_X Q_{AB}}{\kappa} = \frac{\beta\eta v_X N_A N_B}{\kappa} > 0. \tag{47}$$

The maximized welfare gain of a hub with pipeline policy is

$$\Delta^*_{H+b} := \max_{b\geq 0} \Delta_{H+b}(b) = \Delta_H + \frac{\left(\beta\eta v_X Q_{AB}\right)^2}{2\kappa}. \tag{48}$$

Proposition 5 shows that pipeline policy has two properties. First, it is always used when a hub exists, because the marginal benefit of b at $b = 0$ is strictly positive. Second, pipeline policy has purely dynamic value in this benchmark model: it does not improve period-1 local coordination, but it raises period-2 cross-regional renewal and therefore directly offsets the mechanism behind dynamic inefficiency in Section 5.

It is convenient to define the *cross-regional renewal surplus*

$$R_{AB}(N_A, N_B) := \frac{\left(\beta\eta v_X Q_{AB}\right)^2}{2\kappa} = \frac{\beta^2\eta^2 v_X^2 N_A^2 N_B^2}{2\kappa} > 0. \tag{49}$$

Then (48) becomes

$$\Delta^*_{H+b} = \Delta_H + R_{AB}(N_A, N_B). \tag{50}$$

6.2 When Does the Pipeline Rescue the Hub?

Section 4 showed that the hub alone is efficient if and only if

$$F \leq F^*_H(N_A), \qquad F^*_H(N_A) := v_L P_A\left(m^1_A - m^0_A\right)\Lambda_A,$$

where

$$\Lambda_A = (1+\beta) - \beta\delta\big(m_A^1 + m_A^0\big).$$

With pipeline policy, Proposition 5 implies that the relevant fixed-cost threshold becomes

$$F_{H+b}^*(N_A, N_B) := F_H^*(N_A) + R_{AB}(N_A, N_B) = v_L P_A\big(m_A^1 - m_A^0\big)\Lambda_A + \frac{\big(\beta\eta v_X Q_{AB}\big)^2}{2\kappa}. \tag{51}$$

Proposition 6. *A hub combined with optimally chosen cross-regional pipeline policy is socially efficient if and only if*

$$F \le F_{H+b}^*(N_A, N_B). \tag{52}$$

Moreover,

$$F_{H+b}^*(N_A, N_B) - F_H^*(N_A) = R_{AB}(N_A, N_B) > 0, \tag{53}$$

so pipeline policy strictly enlarges the set of parameter values under which a public hub is efficient.

The key comparison is therefore not just between Δ_H and zero, but between Δ_H and $-R_{AB}(N_A, N_B)$. Even when a hub alone is dynamically inefficient, it may become efficient once paired with sufficiently valuable cross-regional renewal.

Proposition 7. *There exists a positive fixed-cost interval for which the hub alone is inefficient but the hub with optimally chosen pipeline policy is efficient if and only if*

$$F_{H+b}^*(N_A, N_B) > \max\{0, F_H^*(N_A)\}. \tag{54}$$

Whenever (54) *holds, the interval is*

$$\max\{0, F_H^*(N_A)\} < F \le F_{H+b}^*(N_A, N_B). \tag{55}$$

Proposition 7 is the central result of this section. It says that the planner may rationally reject a *hub alone* while still approving a *hub plus pipeline policy*. In other words, the relevant policy comparison is not between a hub and no hub in isolation, but between two different institutional designs for the hub.

This logic is especially important when Λ_A is small or negative. In that case, local redundancy erodes much of the hub's stand-alone value. Pipeline policy can nevertheless rescue the hub if the cross-regional renewal channel, summarized by $R_{AB}(N_A, N_B)$, is strong enough.

6.3 Comparative Statics of Pipeline Policy

The next proposition records how the optimal pipeline policy and its welfare contribution vary with the primitive parameters.

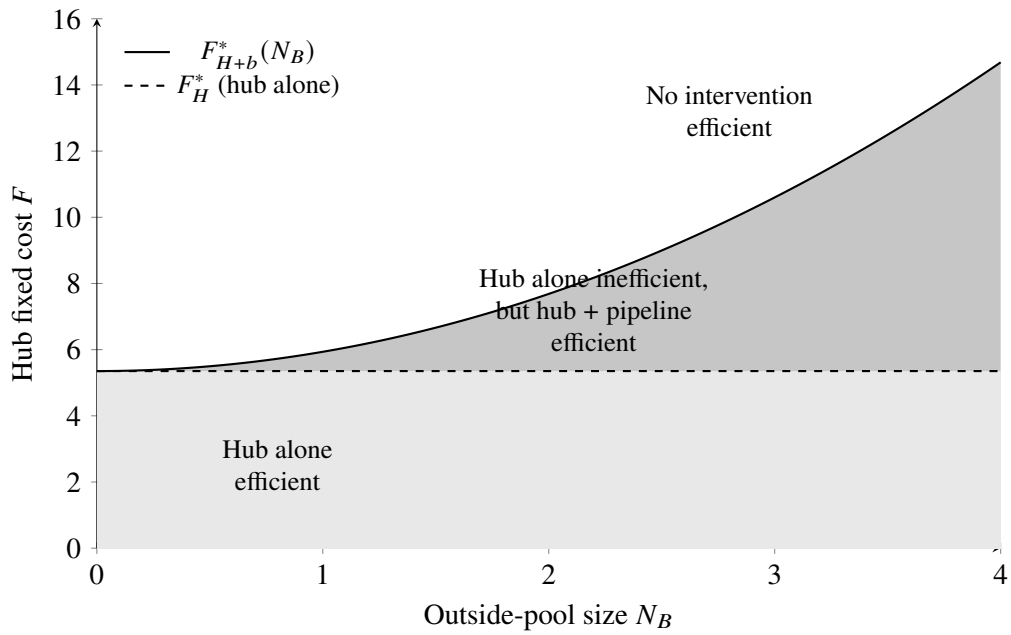

Figure 2: Illustrative policy regions in (N_B, F) space. The dashed line shows the efficiency threshold for a hub alone, F^*_H, while the solid curve shows the threshold for a hub combined with optimally chosen pipeline policy, $F^*_{H+b}(N_B) = F^*_H + (\beta\eta v_X N_A N_B)^2/(2\kappa)$. As the outside pool grows, cross-regional renewal becomes more valuable, enlarging the parameter region in which a hub-plus-pipeline bundle is socially efficient even though a hub alone is not. Parameter values are illustrative: $N_A = 6$, $v_L = 1$, $m^0_A = 0.15$, $m^1_A = 0.55$, $\beta = 0.9$, $\delta = 1.6$, $\eta = 0.2$, $v_X = 1$, and $\kappa = 1$.

Proposition 8. *The optimal pipeline policy b^* and the cross-regional renewal surplus $R_{AB}(N_A, N_B)$ satisfy*

$$\frac{\partial b^*}{\partial N_A} = \frac{\beta\eta v_X N_B}{\kappa} > 0, \tag{56}$$

$$\frac{\partial b^*}{\partial N_B} = \frac{\beta\eta v_X N_A}{\kappa} > 0, \tag{57}$$

$$\frac{\partial b^*}{\partial \eta} = \frac{\beta v_X Q_{AB}}{\kappa} > 0, \tag{58}$$

$$\frac{\partial b^*}{\partial \kappa} = -\frac{\beta\eta v_X Q_{AB}}{\kappa^2} < 0, \tag{59}$$

$$\frac{\partial b^*}{\partial v_X} = \frac{\beta\eta Q_{AB}}{\kappa} > 0, \tag{60}$$

and

$$\frac{\partial R_{AB}(N_A, N_B)}{\partial N_A} = \frac{\beta^2\eta^2 v_X^2 N_A N_B^2}{\kappa} > 0, \tag{61}$$

$$\frac{\partial R_{AB}(N_A, N_B)}{\partial N_B} = \frac{\beta^2\eta^2 v_X^2 N_A^2 N_B}{\kappa} > 0, \tag{62}$$

$$\frac{\partial R_{AB}(N_A, N_B)}{\partial \eta} = \frac{\beta^2\eta v_X^2 Q_{AB}^2}{\kappa} > 0, \tag{63}$$

$$\frac{\partial R_{AB}(N_A, N_B)}{\partial \kappa} = -\frac{\left(\beta\eta v_X Q_{AB}\right)^2}{2\kappa^2} < 0, \tag{64}$$

$$\frac{\partial R_{AB}(N_A, N_B)}{\partial v_X} = \frac{\beta^2\eta^2 v_X Q_{AB}^2}{\kappa} > 0. \tag{65}$$

Proposition 8 shows that cross-regional pipeline policy is more valuable when the target region is larger, when the outside pool is larger, when external linkages generate greater surplus, and when the linkage technology is more effective. It is less valuable when pipeline policy is costly to implement.

Under the minimal benchmark introduced in Section 3, the marginal value of b depends on the size of the cross-regional linkage space Q_{AB} and on the surplus per outside link v_X, but not directly on the local hub matching parameter m_A^1. That feature is deliberate: the benchmark separates the hub's *local coordination role* from the pipeline policy's *cross-regional renewal role*. A richer extension could allow absorptive capacity or anchor participants to make the optimal pipeline intensity depend on local matching strength as well.

6.4 Discussion

The main lesson of this section is not merely that cross-regional pipeline policy is useful. Rather, it is that the socially relevant object is often a *policy bundle*. In the present model, the planner's choice is frequently not between a hub and no hub, but between a hub alone and a hub combined with an outside-linkage mechanism.

This distinction matters because the hub's short-run strength and its long-run weakness are generated by the same source: more intensive local interaction inside region A. A successful hub creates more local

matching, but precisely for that reason it also raises the danger of redundancy and lock-in. Pipeline policy is valuable because it acts on a different margin, namely the supply of cross-regional renewal from region B.

Taken together, Sections 5 and 6 imply that a negative evaluation of a hub-alone policy does not by itself justify abandoning hub policy altogether. It may instead justify redesigning the hub as part of a broader regional strategy that combines local coordination with cross-regional pipeline formation.

7 Extensions

This section develops two extensions that sharpen the interpretation of the baseline model while preserving analytical tractability. The first introduces heterogeneous absorptive capacity among participants in the target region. The second allows digital coordination technologies to substitute for part of the hub's baseline local matching role. In both cases, the central message survives: the welfare value of a public hub depends not only on how much local interaction it generates, but also on whether those interactions remain novel and whether the hub performs a coordination role that cannot be replicated cheaply by other devices.

To streamline notation, define

$$\Delta m_A := m_A^1 - m_A^0 > 0, \qquad \Lambda_A := (1+\beta) - \beta\delta(m_A^1 + m_A^0),$$

so that in the baseline model

$$\Delta_H = v_L P_A \Delta m_A \Lambda_A - F, \qquad b^* = \frac{\beta\eta v_X Q_{AB}}{\kappa}.$$

7.1 Regional Absorptive Capacity and Anchor Actors

The baseline model treats all local participants symmetrically through the common surplus parameter v_L and the common cross-regional surplus parameter v_X. A natural extension allows some local actors in region A to have higher absorptive capacity or stronger boundary-spanning ability. This captures the idea that a region with the same number of participants may nevertheless generate more value if some actors are especially effective at transforming contact into productive collaboration.

Let $\sigma \in [0,1]$ denote the share of *high-absorptive* participants in region A. Suppose that a realized local interaction yields surplus v_L^H if at least one of the two participants is high-absorptive and v_L^L otherwise, where

$$v_L^H > v_L^L > 0.$$

Then the expected surplus per realized local interaction is

$$\bar{v}_L(\sigma) = v_L^L(1-\sigma)^2 + v_L^H\left[1-(1-\sigma)^2\right] = v_L^L + (2\sigma - \sigma^2)(v_L^H - v_L^L). \tag{66}$$

Because

$$\bar{v}_L'(\sigma) = 2(1-\sigma)(v_L^H - v_L^L) > 0 \qquad \text{for } \sigma \in [0,1), \tag{67}$$

a higher share of high-absorptive participants raises the average value of a realized local interaction.

Likewise, suppose that the social surplus from an effective cross-regional link is v_X^H when the region-A

participant involved is high-absorptive and v_X^L otherwise, where

$$v_X^H > v_X^L > 0.$$

Then the expected surplus per effective cross-regional link is

$$\bar{v}_X(\sigma) = v_X^L(1-\sigma) + v_X^H\sigma = v_X^L + \sigma(v_X^H - v_X^L), \tag{68}$$

with

$$\bar{v}_X'(\sigma) = v_X^H - v_X^L > 0. \tag{69}$$

Replacing v_L by $\bar{v}_L(\sigma)$ and v_X by $\bar{v}_X(\sigma)$, the hub-alone welfare gain becomes

$$\Delta_H(\sigma) = \bar{v}_L(\sigma)P_A\Delta m_A\Lambda_A - F, \tag{70}$$

and the optimal pipeline policy conditional on a hub becomes

$$b^*(\sigma) = \frac{\beta\eta\bar{v}_X(\sigma)Q_{AB}}{\kappa}, \tag{71}$$

with corresponding renewal surplus

$$R_{AB}(N_A, N_B; \sigma) = \frac{\left(\beta\eta\bar{v}_X(\sigma)Q_{AB}\right)^2}{2\kappa}. \tag{72}$$

Proposition 9. *Suppose $v_L^H > v_L^L > 0$, $v_X^H > v_X^L > 0$, and $\Lambda_A > 0$. Then:*

1. *The hub-alone welfare gain is increasing in the share of high-absorptive participants:*

$$\frac{\partial\Delta_H(\sigma)}{\partial\sigma} = \bar{v}_L'(\sigma)P_A\Delta m_A\Lambda_A > 0. \tag{73}$$

2. *The optimal pipeline policy is increasing in σ:*

$$\frac{\partial b^*(\sigma)}{\partial\sigma} = \frac{\beta\eta Q_{AB}}{\kappa}\bar{v}_X'(\sigma) > 0. \tag{74}$$

3. *The renewal surplus is increasing in σ:*

$$\frac{\partial R_{AB}(N_A, N_B; \sigma)}{\partial\sigma} = \frac{\beta^2\eta^2 Q_{AB}^2}{\kappa}\bar{v}_X(\sigma)\bar{v}_X'(\sigma) > 0. \tag{75}$$

4. *The efficiency thresholds from Sections 4–6 all shift in favor of the hub as σ rises. In particular, whenever the relevant thresholds are well defined, both $N_H^*(\sigma)$ and $N_{SR}^*(\sigma)$ are decreasing in σ.*

Proposition 9 implies that the effective thickness of a regional ecosystem is not determined by headcount alone. Composition matters. A region with a relatively small number of participants may still support a valuable public hub if enough of those participants have high absorptive capacity or play boundary-spanning roles. Conversely, a larger ecosystem with weak absorptive capacity may fail to justify a hub despite its greater scale.

This extension provides a reduced-form interpretation of anchor firms, strong universities, or capable intermediaries. In the model, such actors raise both the expected value of realized local interaction and the value of cross-regional pipeline formation.

7.2 Digital Substitution for Baseline Local Matching

The baseline model assumes that the hub's matching advantage derives from a reduction in local interaction costs that would otherwise prevail in the region. A natural extension is to allow digital coordination technologies to substitute for part of that role.

Let $o \geq 0$ index the strength of digital coordination tools available in region A. Assume that digital coordination raises the *no-hub* matching rate but leaves the *hub* matching rate unchanged:

$$m_A^0(o) = m_A^0 + \phi o, \qquad \phi > 0, \tag{76}$$

while m_A^1 remains fixed. We restrict attention to the domain

$$0 \leq o < \bar{o} := \frac{m_A^1 - m_A^0}{\phi}, \tag{77}$$

so that

$$0 \leq m_A^0(o) < m_A^1 \leq 1.$$

In this extension, digital tools improve baseline local coordination even without a hub, but they do not replicate the hub's role in organizing or hosting denser in-person interaction.

The hub-alone welfare gain now becomes

$$\Delta_H(o) = v_L P_A \big[m_A^1 - m_A^0(o)\big]\Big[(1+\beta) - \beta\delta\big(m_A^1 + m_A^0(o)\big)\Big] - F. \tag{78}$$

Substituting (76) yields

$$\Delta_H(o) = v_L P_A \big(m_A^1 - m_A^0 - \phi o\big)\Big[(1+\beta) - \beta\delta\big(m_A^1 + m_A^0 + \phi o\big)\Big] - F. \tag{79}$$

The private short-run gain becomes

$$\Pi_H(o) = v_L P_A \big(m_A^1 - m_A^0 - \phi o\big) - F. \tag{80}$$

Proposition 10. *Suppose the domain restriction* (77) *holds and the baseline regularity condition* (6) *is satisfied. Then:*

1. *The short-run gain from the hub is strictly decreasing in digital coordination:*

$$\frac{d\Pi_H(o)}{do} = -\phi v_L P_A < 0. \tag{81}$$

2. *The hub-alone welfare gain is also strictly decreasing in digital coordination:*

$$\frac{d\Delta_H(o)}{do} = -\phi v_L P_A \Big[(1+\beta) - 2\beta\delta m_A^0(o)\Big] < 0. \tag{82}$$

3. *In this reduced form, the optimal pipeline policy conditional on a hub is unchanged:*

$$b^*(o) = \frac{\beta\eta v_X Q_{AB}}{\kappa} = b^*. \tag{83}$$

The intuition is straightforward. Digital coordination tools substitute for the hub's *baseline matching function*: they make it easier for local actors to connect even without a physical hub. As a result, the incremental value of creating the hub falls. However, in this reduced form, digital tools do not alter the hub-side matching intensity m_A^1, nor do they create cross-regional renewal. Hence they reduce the case for hub *creation*, but not the need for pipeline policy once a hub exists.

This extension has two implications. First, the case for a public hub is likely to be weaker in regions where digital coordination is already highly effective. Second, digital substitution does not eliminate the core dynamic problem emphasized in Sections 5 and 6: even if digital tools reduce the need for a hub at the margin, they do not by themselves resolve novelty decay inside a hub-based local network.

7.3 Implications of the Extensions

Taken together, the two extensions refine the baseline theory in opposite directions.

Regional absorptive capacity makes a hub more valuable by increasing the expected surplus from realized interactions and from outside linkages. Digital substitution makes a hub less valuable by reducing the need for a physical coordination device. Yet both extensions leave the main logic intact. A public hub is most likely to be socially desirable when it can generate high-value interactions that would not otherwise occur, and when the resulting interaction pattern remains dynamically productive rather than collapsing into repetitive local exchange.

For the empirical interpretation of the model, the extensions suggest two observable margins beyond simple participant counts. One is the composition of local actors, especially the presence of anchor or highly absorptive participants. The other is the availability of alternative coordination technologies, especially digital tools that may replicate part of the hub's function without resolving the hub's dynamic novelty problem.

8 Empirical Predictions and Regional Policy Interpretation

The model is intended as a mechanism-based framework rather than as a literal description of any single hub program. Its empirical value lies in clarifying which observable margins correspond to the theoretical objects introduced above, and in identifying predictions that distinguish a hub's visible short-run success from its broader regional welfare effects. For the present paper, the most important implication is that the relevant empirical comparison is often not simply between *hub* and *no hub*, but between a *hub alone* and a *hub embedded in a cross-regional renewal strategy*.

A convenient starting point is the decomposition developed in Section 5. The hub's private short-run appeal is summarized by

$$\Pi_H = S_A - F, \qquad S_A := v_L P_A(m_A^1 - m_A^0),$$

while its social value as a stand-alone policy is

$$\Delta_H = \Pi_H + D_A, \qquad D_A := \beta v_L P_A(m_A^1 - m_A^0)\Big[1 - \delta(m_A^1 + m_A^0)\Big].$$

With pipeline policy, social value becomes

$$\Delta^*_{H+b} = \Pi_H + D_A + R_{AB}(N_A, N_B),$$

where

$$R_{AB}(N_A, N_B) = \frac{(\beta\eta v_X Q_{AB})^2}{2\kappa} > 0.$$

This decomposition immediately suggests what should be measured empirically. A good evaluation should not stop at visible activity generated by the hub in region A; it should also track whether local interaction becomes repetitive over time and whether policy succeeds in creating cross-regional renewal through outside linkages.

8.1 Mapping Theory to Observable Margins

Table 1 summarizes a parsimonious correspondence between the model's theoretical objects and plausible empirical proxies.

Table 1: Illustrative Mapping from Theoretical Objects to Empirical Proxies

Theoretical object	Empirical interpretation	Illustrative proxies
Local matching gain $m^1_A - m^0_A$	Immediate coordination effect of the hub inside region A	Interaction volume, first-time meetings, new project starts, participant turnover
Dynamic redundancy δ	Speed at which local interaction becomes repetitive or locked in	New ties share, repeated collaborator ratio, decline in participant turnover, network modularity or concentration of repeated ties
Cross-regional renewal b and $q(b)$	Outside-linkage formation and extra-regional brokerage	Outside-region collaborator share, invited outside participants, co-patenting/co-application novelty, external mentor or university linkages
Absorptive capacity / anchor presence	Capacity to convert contact into productive regional surplus	Anchor participation share, lead-firm or university presence, share of boundary-spanning intermediaries

The purpose of Table 1 is not to impose a single empirical design. Rather, it highlights that the model's core variables are observable at least in approximate form. In particular, the model suggests that a credible empirical evaluation should include not only gross activity measures, but also indicators of *novelty*, *repetition*, and *outside connectivity*.

8.2 Three Testable Predictions

The model yields three testable predictions that are especially relevant for regional policy evaluation.

Prediction 1. Hub creation raises initial interaction volume, but does not necessarily sustain the share of new ties. The baseline model implies an immediate local coordination gain,

$$B_1(1) - B_1(0) = v_L P_A(m_A^1 - m_A^0) > 0.$$

Thus, after hub establishment, one should expect an increase in visible local activity in region A: more meetings, more matches, more projects, or more observed interaction volume. This is the mechanism behind the private short-run criterion Π_H.

However, Section 5 also shows that the hub's dynamic correction is

$$D_A = \beta v_L P_A(m_A^1 - m_A^0)\Big[1 - \delta(m_A^1 + m_A^0)\Big].$$

When local redundancy is strong, this term becomes small or negative. In empirical terms, the model therefore predicts that early increases in interaction volume need not translate into persistently high novelty. Instead, one should expect some hubs to exhibit rising *repeated collaborator ratios* or falling *new ties shares* over time, especially where the regional participant pool is narrow and the same actors repeatedly interact.

The empirical implication is straightforward: evaluations based only on attendance, occupancy, or total interactions are incomplete. A more theory-consistent design should track whether first-time collaborations continue to form after the hub's initial launch period. In the model's language, high short-run activity is evidence about S_A, but not by itself about D_A.

Prediction 2. Regions with a hub plus pipeline policy should exhibit more persistent outside collaboration and higher novelty than otherwise similar hub-only regions. Section 6 shows that conditional on a hub, the planner chooses

$$b^* = \frac{\beta\eta v_X Q_{AB}}{\kappa} > 0,$$

and that the resulting incremental renewal surplus is

$$R_{AB}(N_A, N_B) = \frac{(\beta\eta v_X Q_{AB})^2}{2\kappa} > 0.$$

The interpretation is that pipeline policy acts on a different margin from the hub itself. The hub raises local coordination inside region A, whereas the pipeline policy raises cross-regional connections between region A and the outside pool B.

This implies a testable contrast between *hub-only* regions and *hub-plus-pipeline* regions. Relative to otherwise similar hub-only regions, regions that combine the hub with active outside-linkage policy should display a higher *outside-region collaborator share*, greater *co-patenting or co-application novelty*, and a more persistent flow of new collaborative combinations in the medium run. They should also be less likely to exhibit the stagnation pattern described in Prediction 1.

From a policy perspective, this prediction is central. The theory does not merely claim that outside linkages are useful in general. It claims that they are especially valuable when a hub has already increased local interaction intensity and therefore increased the danger of repetition. Empirically, the relevant comparison is therefore not simply before versus after the hub, but whether hub effects differ systematically according to the strength of extra-regional linkage formation.

Prediction 3. The same hub policy generates larger welfare gains in regions with higher absorptive capacity. The heterogeneous-participant extension implies that composition matters in addition to scale. A region with the same number of participants may nevertheless generate higher surplus if a larger share of those participants are especially effective at transforming contact into productive collaboration. In practice, this corresponds to the presence of anchor firms, strong universities, capable intermediaries, or other actors that raise the expected value of realized interactions.

The empirical implication is that the effect of hub policy should be stronger in regions with greater *absorptive capacity*. Observable proxies include *anchor participation share*, the presence of universities or lead firms, or the share of participants with boundary-spanning roles. Relative to observationally similar hubs in low-capacity settings, hubs in high-capacity settings should exhibit higher persistence of novel collaborations, stronger project conversion, and larger medium-run gains from outside-linkage policy.

This prediction matters because it sharpens the interpretation of ecosystem thickness. The theory does not say that headcount alone determines efficiency. Rather, it suggests that the effective thickness of a region depends jointly on the number of plausible participants and on the composition of the participant pool. In empirical work, this means that a hub in a smaller region should not automatically be expected to perform poorly if that region contains sufficiently strong anchor actors or unusually capable intermediaries.

8.3 Implications for Empirical Design

Taken together, the three predictions imply that a theory-consistent empirical evaluation of hub policy should be dynamic, network-aware, and explicitly regional.

First, evaluation should be *dynamic*. Early activity measures capture the short-run coordination term S_A, but do not reveal whether novelty persists. At a minimum, researchers should compare initial interaction gains with the subsequent evolution of new ties share, repeated collaborator ratio, and participant turnover.

Second, evaluation should be *network-aware*. The model is fundamentally about the difference between creating interactions and creating *nonrepetitive* interactions. Measures such as repeated collaborator ratio, outside-region collaborator share, network modularity, and the novelty content of co-patenting or co-application patterns are therefore better aligned with the model than aggregate attendance alone.

Third, evaluation should be *explicitly regional*. The two-region formulation implies that the outside pool is not a residual background feature, but a policy-relevant object. This means that data on external participants, outside mentors, nonlocal co-applicants, or interregional partner flows are not peripheral controls; they are central outcome variables for judging whether a hub is functioning as part of a broader regional strategy.

Overall, the empirical agenda suggested by the model is therefore sharper than a conventional utilization-based evaluation. The critical question is not simply whether the hub generated more activity, but whether it generated activity that remained novel and whether policy design created enough outside renewal to prevent local coordination from hardening into local repetition.

9 Conclusion

This paper has developed a two-region dynamic framework for evaluating public co-creation hubs in thin regional ecosystems. The model distinguishes between two margins of regional policy. The first is *local coordination*: a public hub raises the rate at which potentially valuable local interactions are realized within

a target region. The second is *cross-regional renewal*: a complementary pipeline policy creates outside linkages that bring additional collaborators, ideas, and opportunities into the regional system. The paper's central claim is that these two margins should not be evaluated separately.

Three conclusions follow. First, a hub is socially efficient as a stand-alone intervention only if its local coordination gains are large enough to offset fixed cost and the dynamic redundancy created by repeated local interaction. The welfare case for a hub therefore depends not only on ecosystem thickness and the hub's matching effect, but also on the pace at which novelty is exhausted inside the local network. Second, short-run success is not equivalent to social efficiency. A hub may generate visible early activity and still be socially undesirable once repeated local ties and dynamic novelty loss are taken into account. Third, the relevant policy comparison is often not between a hub and no hub, but between a hub alone and a hub embedded in a broader regional strategy. A stand-alone hub may fail a welfare test even though a hub combined with cross-regional pipeline formation is socially efficient.

The extensions reinforce these conclusions. Participant composition matters in addition to headcount, so the value of a hub depends not only on how many actors a region contains, but also on whether those actors include highly absorptive or boundary-spanning participants. Conversely, digital coordination technologies weaken the case for physical hub creation when they substitute for baseline local matching, although they do not by themselves eliminate the problem of dynamic redundancy within a narrow regional network.

The paper also points to a more demanding empirical agenda. A theory-consistent evaluation of hub policy should not rely only on attendance, occupancy, or other utilization-based indicators. It should also examine whether new ties continue to form, whether repeated collaborator ratios rise over time, whether outside-region collaboration expands, and whether regions with stronger absorptive capacity derive larger gains from the same policy design. In this sense, the paper suggests that empirical work on hubs should be dynamic, network-aware, and explicitly regional.

The framework is intentionally stylized, and several extensions remain open. A richer model could allow for endogenous participant entry, repeated periods beyond the current two-period setting, strategic interaction across jurisdictions, or explicit network formation rather than reduced-form matching. An empirical extension could bring the theory closer to administrative or project-level data on collaboration networks, outside partners, and participant turnover.

Even in its simplified form, however, the model delivers a clear message for regional policy. In thin regional ecosystems, public hubs should not be judged as stand-alone venues that either succeed or fail according to visible early activity. They should be evaluated as components of a broader place-based strategy that combines local buzz with cross-regional renewal.

References

References

Agrawal, A., and I. Cockburn. 2003. The anchor tenant hypothesis: Exploring the role of large, local, R&D-intensive firms in regional innovation systems. *International Journal of Industrial Organization* 21(9): 1227–1253. https://doi.org/10.1016/S0167-7187(03)00081-X.

Asheim, B. T., H. Lawton Smith, and C. Oughton. 2011. Regional innovation systems: Theory, empirics and policy. *Regional Studies* 45(7): 875–891. https://doi.org/10.1080/00343404.2011.596701.

Bathelt, H., A. Malmberg, and P. Maskell. 2004. Clusters and knowledge: Local buzz, global pipelines and the process of knowledge creation. *Progress in Human Geography* 28(1): 31–56. https://doi.org/10.1191/0309132504ph469oa.

Boschma, R. A. 2005. Proximity and innovation: A critical assessment. *Regional Studies* 39(1): 61–74. https://doi.org/10.1080/0034340052000320887.

Chaminade, C., M. Bellandi, M. Plechero, and E. Santini. 2019. Understanding processes of path renewal and creation in thick specialized regional innovation systems: Evidence from two textile districts in Italy and Sweden. *European Planning Studies* 27(10): 1978–1994. https://doi.org/10.1080/09654313.2019.1610727.

Coenen, L., B. Asheim, M. M. Bugge, and S. J. Herstad. 2017. Advancing regional innovation systems: What does evolutionary economic geography bring to the policy table? *Environment and Planning C: Politics and Space* 35(4): 600–620. https://doi.org/10.1177/0263774X16646583.

Cohen, W. M., and D. A. Levinthal. 1990. Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly* 35(1): 128–152. https://doi.org/10.2307/2393553.

Cooke, P., M. Gómez Uranga, and G. Etxebarria. 1997. Regional innovation systems: Institutional and organisational dimensions. *Research Policy* 26(4–5): 475–491. https://doi.org/10.1016/S0048-7333(97)00025-5.

Duranton, G., and D. Puga. 2004. Micro-foundations of urban agglomeration economies. In *Handbook of Regional and Urban Economics*, vol. 4, edited by J. V. Henderson and J.-F. Thisse, 2063–2117. Amsterdam: Elsevier. https://doi.org/10.1016/S1574-0080(04)80005-1.

Jaffe, A. B., M. Trajtenberg, and R. Henderson. 1993. Geographic localization of knowledge spillovers as evidenced by patent citations. *The Quarterly Journal of Economics* 108(3): 577–598. https://doi.org/10.2307/2118401.

Kline, P., and E. Moretti. 2014a. People, places, and public policy: Some simple welfare economics of local economic development programs. *Annual Review of Economics* 6: 629–662. https://doi.org/10.1146/annurev-economics-080213-041024.

Kline, P., and E. Moretti. 2014b. Local economic development, agglomeration economies, and the big push: 100 years of evidence from the Tennessee Valley Authority. *The Quarterly Journal of Economics* 129(1): 275–331. https://doi.org/10.1093/qje/qjt034.

Merrell, I., J. Phillipson, M. Gorton, and P. Cowie. 2022. Enterprise hubs as a mechanism for local economic development in rural areas. *Journal of Rural Studies* 93: 81–91. https://doi.org/10.1016/j.jrurstud.2022.05.016.

Merrell, I., F. Rowe, P. Cowie, and M. Gkartzios. 2021. 'Honey Pot' rural enterprise hubs as micro-clusters: Exploring their role in creativity-led rural development. *Local Economy* 36(7–8): 589–605. https://doi.org/10.1177/02690942221085498.

Neumark, D., and H. Simpson. 2015. Place-based policies. In *Handbook of Regional and Urban Economics*, vol. 5, edited by G. Duranton, J. V. Henderson, and W. C. Strange, 1197–1287. Amsterdam: Elsevier. https://doi.org/10.1016/B978-0-444-59531-7.00018-1.

OECD. 2025. *Rural Innovation Pathways: Connecting People, Places and Ideas*. OECD Rural Studies. Paris: OECD Publishing. https://doi.org/10.1787/c86de0f4-en.

Roche, M. P., A. Oettl, and C. Catalini. 2024. Proximate (co-)working: Knowledge spillovers and social interactions. *Management Science* 70(12): 8245–8264. https://doi.org/10.1287/mnsc.2022.03555.

Rosenthal, S. S., and W. C. Strange. 2004. Evidence on the nature and sources of agglomeration economies. In *Handbook of Regional and Urban Economics*, vol. 4, edited by J. V. Henderson and J.-F. Thisse, 2119–2171. Amsterdam: Elsevier. https://doi.org/10.1016/S1574-0080(04)80006-3.

Storper, M., and A. J. Venables. 2004. Buzz: Face-to-face contact and the urban economy. *Journal of Economic Geography* 4(4): 351–370. https://doi.org/10.1093/jeg/4.4.351.

T"odtling, F., and M. Trippl. 2005. One size fits all? Towards a differentiated regional innovation policy approach. *Research Policy* 34(8): 1203–1219. https://doi.org/10.1016/j.respol.2005.01.018.

Torre, A., and A. Rallet. 2005. Proximity and localization. *Regional Studies* 39(1): 47–59. https://doi.org/10.1080/0034340052000320842.

Vogl, T., and M. Akhavan. 2022. A systematic literature review of the effects of coworking spaces on the socio-cultural and economic conditions in peripheral and rural areas. *Journal of Property Investment & Finance* 40(5): 465–478. https://doi.org/10.1108/JPIF-12-2021-0108.

A Proofs for Section 4

This appendix collects the algebra omitted from Section 4.

A.1 Preliminaries

Recall from Section 3 that

$$P_A = \frac{N_A(N_A-1)}{2}, \qquad Q_{AB} = N_A N_B, \qquad m_A(h) = m_A^0 + \big(m_A^1 - m_A^0\big)h,$$

with

$$0 \le m_A^0 < m_A^1 \le 1.$$

In the benchmark case without pipeline policy ($b = 0$), welfare is

$$W(0,0) = v_L P_A m_A^0 + \beta\Big\{v_L P_A m_A^0\big(1-\delta m_A^0\big) + v_X Q_{AB} q_0\Big\}, \tag{84}$$

and

$$W(1,0) = v_L P_A m_A^1 + \beta\Big\{v_L P_A m_A^1\big(1-\delta m_A^1\big) + v_X Q_{AB} q_0\Big\} - F. \tag{85}$$

Hence the net welfare gain from a hub alone is

$$\begin{aligned}\Delta_H &:= W(1,0) - W(0,0)\\ &= v_L P_A\Big[m_A^1 - m_A^0 + \beta\big(m_A^1(1-\delta m_A^1) - m_A^0(1-\delta m_A^0)\big)\Big] - F.\end{aligned} \tag{86}$$

The cross-regional baseline term cancels exactly:

$$\beta v_X Q_{AB} q_0 - \beta v_X Q_{AB} q_0 = 0.$$

Thus the hub-alone comparison depends only on local primitives.

Now,

$$\begin{aligned} m_A^1(1-\delta m_A^1) - m_A^0(1-\delta m_A^0) &= (m_A^1 - m_A^0) - \delta\big((m_A^1)^2 - (m_A^0)^2\big) \\ &= (m_A^1 - m_A^0)\big[1 - \delta(m_A^1 + m_A^0)\big]. \end{aligned} \tag{87}$$

Substituting (87) into (86) yields

$$\Delta_H = v_L P_A (m_A^1 - m_A^0)\Big[(1+\beta) - \beta\delta(m_A^1 + m_A^0)\Big] - F. \tag{88}$$

Define

$$\Lambda_A := (1+\beta) - \beta\delta(m_A^1 + m_A^0), \tag{89}$$

so that

$$\Delta_H = v_L P_A (m_A^1 - m_A^0)\Lambda_A - F. \tag{90}$$

A.2 Proof of Proposition 1

We prove each part in turn.

Part 1: the case $\Lambda_A \le 0$. Since $v_L > 0$, $P_A > 0$ for all $N_A \ge 2$, and $m_A^1 - m_A^0 > 0$, the product

$$v_L P_A (m_A^1 - m_A^0)\Lambda_A$$

is weakly negative whenever $\Lambda_A \le 0$. Therefore, from (90),

$$\Delta_H = v_L P_A (m_A^1 - m_A^0)\Lambda_A - F < 0$$

for every $F > 0$. Hence the hub is never socially efficient.

Part 2: the case $\Lambda_A > 0$. Under $\Lambda_A > 0$, the condition $\Delta_H \ge 0$ is equivalent to

$$v_L P_A (m_A^1 - m_A^0)\Lambda_A \ge F.$$

Since all terms except P_A are strictly positive, this can be rearranged as

$$P_A \ge \frac{F}{v_L (m_A^1 - m_A^0)\Lambda_A}. \tag{91}$$

This proves (21).

Using $P_A = N_A(N_A - 1)/2$, (91) becomes

$$\frac{N_A(N_A-1)}{2} \geq \frac{F}{v_L(m_A^1 - m_A^0)\Lambda_A},$$

or equivalently

$$N_A^2 - N_A - \frac{2F}{v_L(m_A^1 - m_A^0)\Lambda_A} \geq 0.$$

The positive root of the corresponding quadratic equation is

$$\frac{1}{2}\left(1 + \sqrt{1 + \frac{8F}{v_L(m_A^1 - m_A^0)\Lambda_A}}\right).$$

Hence, treating N_A as continuous, the hub is efficient if and only if

$$N_A \geq N_H^* := \frac{1}{2}\left(1 + \sqrt{1 + \frac{8F}{v_L(m_A^1 - m_A^0)\Lambda_A}}\right), \tag{92}$$

which is exactly (22). For integer N_A, replace N_H^* by its ceiling. This completes the proof. □

A.3 Derivation of the Equivalent Threshold Characterizations

Starting from (90), the condition $\Delta_H \geq 0$ is equivalent to

$$F \leq v_L P_A(m_A^1 - m_A^0)\Lambda_A.$$

Thus the critical fixed cost is

$$F_H^*(N_A) := v_L P_A(m_A^1 - m_A^0)\Lambda_A, \tag{93}$$

which is (23).

Next, solve for the critical redundancy rate. Rewrite $\Delta_H \geq 0$ as

$$v_L P_A(m_A^1 - m_A^0)\Big[(1+\beta) - \beta\delta(m_A^1 + m_A^0)\Big] \geq F.$$

Dividing by the positive term $v_L P_A(m_A^1 - m_A^0)$ gives

$$(1+\beta) - \beta\delta(m_A^1 + m_A^0) \geq \frac{F}{v_L P_A(m_A^1 - m_A^0)}.$$

Hence

$$\beta\delta(m_A^1 + m_A^0) \leq (1+\beta) - \frac{F}{v_L P_A(m_A^1 - m_A^0)},$$

so

$$\begin{aligned}\delta &\leq \frac{1+\beta}{\beta(m_A^1 + m_A^0)} - \frac{F}{\beta v_L P_A(m_A^1 - m_A^0)(m_A^1 + m_A^0)} \\ &=: \delta_H^*(N_A),\end{aligned} \tag{94}$$

which is (24).

A.4 Proof of Proposition 2

Throughout this proof we use

$$\Delta_H = v_L P_A(m_A^1 - m_A^0)\Lambda_A - F, \qquad \Lambda_A = (1+\beta) - \beta\delta(m_A^1 + m_A^0).$$

Derivative with respect to N_A. Treating N_A as continuous,

$$P_A = \frac{N_A(N_A-1)}{2} \quad \Rightarrow \quad \frac{\partial P_A}{\partial N_A} = \frac{2N_A - 1}{2}.$$

Hence

$$\begin{aligned}\frac{\partial \Delta_H}{\partial N_A} &= v_L \frac{\partial P_A}{\partial N_A}(m_A^1 - m_A^0)\Lambda_A \\ &= \frac{v_L}{2}(2N_A - 1)(m_A^1 - m_A^0)\Lambda_A, \end{aligned} \tag{95}$$

which is (25). If $\Lambda_A > 0$, then $\partial\Delta_H/\partial N_A > 0$ because $v_L > 0$, $2N_A - 1 > 0$, and $m_A^1 - m_A^0 > 0$.

For integer N_A, the same monotonicity follows from forward differences:

$$\begin{aligned}\Delta_H(N_A+1) - \Delta_H(N_A) &= v_L\big(P_A(N_A+1) - P_A(N_A)\big)(m_A^1 - m_A^0)\Lambda_A \\ &= v_L N_A(m_A^1 - m_A^0)\Lambda_A > 0 \quad \text{if } \Lambda_A > 0. \end{aligned} \tag{96}$$

Derivative with respect to F. From (90),

$$\frac{\partial \Delta_H}{\partial F} = -1 < 0, \tag{97}$$

which is (26).

Derivative with respect to δ. Differentiating $\Lambda_A = (1+\beta) - \beta\delta(m_A^1 + m_A^0)$ gives

$$\frac{\partial \Lambda_A}{\partial \delta} = -\beta(m_A^1 + m_A^0).$$

Therefore

$$\begin{aligned}\frac{\partial \Delta_H}{\partial \delta} &= v_L P_A(m_A^1 - m_A^0)\frac{\partial \Lambda_A}{\partial \delta} \\ &= -\beta v_L P_A(m_A^1 - m_A^0)(m_A^1 + m_A^0) < 0, \end{aligned} \tag{98}$$

which is (27).

Derivative with respect to m_A^1. Starting from

$$\Delta_H = v_L P_A (m_A^1 - m_A^0)\Big[(1+\beta) - \beta\delta(m_A^1 + m_A^0)\Big] - F,$$

differentiate with respect to m_A^1:

$$\begin{aligned}\frac{\partial \Delta_H}{\partial m_A^1} &= v_L P_A \left(\Big[(1+\beta) - \beta\delta(m_A^1 + m_A^0)\Big] + (m_A^1 - m_A^0)(-\beta\delta)\right) \\ &= v_L P_A \Big[(1+\beta) - 2\beta\delta m_A^1\Big]. \end{aligned} \tag{99}$$

This is exactly (28).

To verify the sign, note that the regularity condition (6) implies

$$1 - \delta m_A^1 > 0 \quad \Longrightarrow \quad \delta m_A^1 < 1.$$

Hence

$$2\beta\delta m_A^1 < 2\beta.$$

Since $\beta \in (0, 1]$, we have $1 + \beta \geq 2\beta$, with equality only when $\beta = 1$. Therefore,

$$(1+\beta) - 2\beta\delta m_A^1 > (1+\beta) - 2\beta = 1 - \beta \geq 0,$$

where the strict inequality follows from $\delta m_A^1 < 1$. Thus

$$\frac{\partial \Delta_H}{\partial m_A^1} > 0. \tag{100}$$

Derivative with respect to m_A^0. Similarly,

$$\begin{aligned}\frac{\partial \Delta_H}{\partial m_A^0} &= v_L P_A \left(-\Big[(1+\beta) - \beta\delta(m_A^1 + m_A^0)\Big] + (m_A^1 - m_A^0)(-\beta\delta)\right) \\ &= -\, v_L P_A \Big[(1+\beta) - 2\beta\delta m_A^0\Big]. \end{aligned} \tag{101}$$

This is exactly (29).

Because $m_A^0 < m_A^1$ and $\delta m_A^1 < 1$, we also have $\delta m_A^0 < 1$. Repeating the same sign argument as above gives

$$(1+\beta) - 2\beta\delta m_A^0 > 0,$$

so

$$\frac{\partial \Delta_H}{\partial m_A^0} < 0. \tag{102}$$

This proves Proposition 2. □

B Proofs for Section 5

This appendix provides the proofs omitted from Section 5.

B.1 Proof of Proposition 3

By definition,

$$\Pi_H = \left[B_1(1) - F\right] - B_1(0).$$

Using (4),

$$B_1(1) = v_L P_A m_A^1, \qquad B_1(0) = v_L P_A m_A^0.$$

Hence

$$\begin{aligned}\Pi_H &= v_L P_A m_A^1 - F - v_L P_A m_A^0 \\ &= v_L P_A (m_A^1 - m_A^0) - F,\end{aligned} \tag{103}$$

which is (31).

Therefore,

$$\Pi_H > 0 \quad \Longleftrightarrow \quad F < v_L P_A (m_A^1 - m_A^0),$$

which proves (32).

To derive the threshold in terms of N_A, substitute

$$P_A = \frac{N_A(N_A - 1)}{2}$$

into (32):

$$\frac{N_A(N_A - 1)}{2} > \frac{F}{v_L(m_A^1 - m_A^0)}.$$

Equivalently,

$$N_A^2 - N_A - \frac{2F}{v_L(m_A^1 - m_A^0)} > 0.$$

The positive root of the associated quadratic equation is

$$\frac{1}{2}\left(1 + \sqrt{1 + \frac{8F}{v_L(m_A^1 - m_A^0)}}\right).$$

Hence, treating N_A as continuous, private support obtains if and only if

$$N_A \geq N_{SR}^* := \frac{1}{2}\left(1 + \sqrt{1 + \frac{8F}{v_L(m_A^1 - m_A^0)}}\right),$$

with the usual ceiling if N_A must be an integer. □

B.2 Derivation of the Private–Social Wedge

Recall the definitions

$$S_A := v_L P_A(m_A^1 - m_A^0),$$

$$D_A := \beta v_L P_A(m_A^1 - m_A^0)\big[1 - \delta(m_A^1 + m_A^0)\big],$$

and

$$S_{AB}(b) := \beta\eta v_X Q_{AB} b - \frac{\kappa}{2} b^2.$$

Since

$$\Pi_H = S_A - F$$

and

$$\Delta_H = v_L P_A(m_A^1 - m_A^0)\Big[(1+\beta) - \beta\delta(m_A^1 + m_A^0)\Big] - F,$$

we obtain

$$\begin{aligned}\Delta_H - \Pi_H &= \beta v_L P_A(m_A^1 - m_A^0)\big[1 - \delta(m_A^1 + m_A^0)\big] \\ &= D_A. \end{aligned} \tag{104}$$

Hence

$$\Delta_H = \Pi_H + D_A,$$

which is (37).

Likewise, from Section 6,

$$\Delta_{H+b}(b) = \Delta_H + S_{AB}(b),$$

so

$$\Delta_{H+b}(b) = \Pi_H + D_A + S_{AB}(b),$$

which is (38).

Finally, using the optimal pipeline policy from Section 6,

$$R_{AB}(N_A, N_B) = \frac{(\beta\eta v_X Q_{AB})^2}{2\kappa},$$

we obtain

$$\Delta_{H+b}^* = \Pi_H + D_A + R_{AB}(N_A, N_B),$$

which is (40).

B.3 Proof of Proposition 4

Recall

$$\Pi_H = S_A - F, \qquad \Delta_H = S_A + D_A - F, \qquad \Delta_{H+b}^* = S_A + D_A + R_{AB}(N_A, N_B) - F.$$

Hence

$$F_H^*(N_A) = S_A + D_A, \qquad F_{H+b}^*(N_A, N_B) = S_A + D_A + R_{AB}(N_A, N_B).$$

Part 1: Private over-support of a hub alone. If

$$\delta(m_A^1 + m_A^0) > 1,$$

then

$$1 - \delta(m_A^1 + m_A^0) < 0,$$

so by definition,

$$D_A = \beta v_L P_A (m_A^1 - m_A^0)\left[1 - \delta(m_A^1 + m_A^0)\right] < 0.$$

Therefore

$$\Delta_H = \Pi_H + D_A < \Pi_H.$$

Next, the hub is privately supported under the short-run rule if and only if

$$\Pi_H > 0 \quad \Longleftrightarrow \quad F < S_A.$$

The hub is socially undesirable as a stand-alone policy if and only if

$$\Delta_H < 0 \quad \Longleftrightarrow \quad F > F_H^*(N_A).$$

Combining these two conditions gives

$$\max\{0, F_H^*(N_A)\} < F < S_A.$$

It remains to show that this interval is nonempty. Since $D_A < 0$,

$$F_H^*(N_A) = S_A + D_A < S_A.$$

Hence

$$\max\{0, F_H^*(N_A)\} < S_A,$$

so the interval is nonempty. This proves (42).

Part 2: Private under-support of a hub-plus-pipeline bundle. If

$$D_A + R_{AB}(N_A, N_B) > 0,$$

then

$$F_{H+b}^*(N_A, N_B) = S_A + D_A + R_{AB}(N_A, N_B) > S_A.$$

The hub is not privately supported under the short-run rule if and only if

$$\Pi_H \leq 0 \quad \Longleftrightarrow \quad F \geq S_A.$$

The hub-plus-pipeline bundle is socially desirable if and only if

$$\Delta_{H+b}^* > 0 \quad \Longleftrightarrow \quad F < F_{H+b}^*(N_A, N_B).$$

Combining these gives

$$S_A \leq F < F^*_{H+b}(N_A, N_B),$$

which is exactly (44). Because

$$F^*_{H+b}(N_A, N_B) > S_A,$$

this interval is nonempty. This completes the proof. □

C Proofs for Section 6

This appendix provides the proofs omitted from Section 6.

C.1 Proof of Proposition 5

Conditional on $h = 1$, the planner chooses $b \geq 0$ to maximize

$$\Delta_{H+b}(b) = \Delta_H + \beta\eta v_X Q_{AB} b - \frac{\kappa}{2} b^2.$$

Differentiating with respect to b,

$$\frac{d\Delta_{H+b}(b)}{db} = \beta\eta v_X Q_{AB} - \kappa b. \tag{105}$$

The second derivative is

$$\frac{d^2\Delta_{H+b}(b)}{db^2} = -\kappa < 0, \tag{106}$$

so the problem is strictly concave and the first-order condition is sufficient.

Setting (105) equal to zero gives

$$\beta\eta v_X Q_{AB} - \kappa b = 0,$$

hence

$$b^* = \frac{\beta\eta v_X Q_{AB}}{\kappa} = \frac{\beta\eta v_X N_A N_B}{\kappa}. \tag{107}$$

Because $\beta > 0$, $\eta > 0$, $v_X > 0$, $Q_{AB} = N_A N_B > 0$, and $\kappa > 0$, we have $b^* > 0$. Thus the nonnegativity constraint does not bind.

Substituting (107) into $\Delta_{H+b}(b)$,

$$\begin{aligned}
\Delta^*_{H+b} &= \Delta_H + \beta\eta v_X Q_{AB}\left(\frac{\beta\eta v_X Q_{AB}}{\kappa}\right) - \frac{\kappa}{2}\left(\frac{\beta\eta v_X Q_{AB}}{\kappa}\right)^2 \\
&= \Delta_H + \frac{(\beta\eta v_X Q_{AB})^2}{\kappa} - \frac{1}{2}\frac{(\beta\eta v_X Q_{AB})^2}{\kappa} \\
&= \Delta_H + \frac{(\beta\eta v_X Q_{AB})^2}{2\kappa},
\end{aligned} \tag{108}$$

which is (48). This proves Proposition 5. □

C.2 Proof of Proposition 6

By Proposition 5,

$$\Delta^*_{H+b} = \Delta_H + \frac{(\beta\eta v_X Q_{AB})^2}{2\kappa}.$$

From Section 4,

$$\Delta_H = v_L P_A(m^1_A - m^0_A)\Lambda_A - F, \qquad \Lambda_A = (1+\beta) - \beta\delta(m^1_A + m^0_A).$$

Hence

$$\begin{aligned}\Delta^*_{H+b} &= v_L P_A(m^1_A - m^0_A)\Lambda_A + \frac{(\beta\eta v_X Q_{AB})^2}{2\kappa} - F \\ &= F^*_{H+b}(N_A, N_B) - F, \end{aligned} \tag{109}$$

where

$$F^*_{H+b}(N_A, N_B) := v_L P_A(m^1_A - m^0_A)\Lambda_A + \frac{(\beta\eta v_X Q_{AB})^2}{2\kappa}. \tag{110}$$

Therefore,

$$\Delta^*_{H+b} \geq 0 \quad \Longleftrightarrow \quad F \leq F^*_{H+b}(N_A, N_B),$$

which proves (52).

Next,

$$F^*_H(N_A) = v_L P_A(m^1_A - m^0_A)\Lambda_A,$$

so

$$\begin{aligned}F^*_{H+b}(N_A, N_B) - F^*_H(N_A) &= \frac{(\beta\eta v_X Q_{AB})^2}{2\kappa} \\ &=: R_{AB}(N_A, N_B) > 0, \end{aligned} \tag{111}$$

which is exactly (53). Thus pipeline policy strictly enlarges the efficient set. This proves Proposition 6. □

C.3 Proof of Proposition 7

A hub alone is inefficient if and only if

$$\Delta_H < 0.$$

Since

$$\Delta_H = F^*_H(N_A) - F,$$

this is equivalent to

$$F > F^*_H(N_A). \tag{112}$$

A hub with optimally chosen pipeline policy is efficient if and only if

$$\Delta^*_{H+b} \geq 0,$$

which, by Proposition 6, is equivalent to

$$F \leq F^*_{H+b}(N_A, N_B). \tag{113}$$

Because fixed cost is assumed positive, the admissible set is $F > 0$. Combining $F > 0$, (112), and (113), the set of fixed costs for which the hub alone is inefficient but the hub with pipeline policy is efficient is

$$\max\{0, F^*_H(N_A)\} < F \leq F^*_{H+b}(N_A, N_B).$$

This set is nonempty if and only if its upper bound exceeds its lower bound, i.e.

$$F^*_{H+b}(N_A, N_B) > \max\{0, F^*_H(N_A)\},$$

which proves (54). The interval itself is exactly (55). This proves Proposition 7. □

C.4 Proof of Proposition 8

Recall

$$b^* = \frac{\beta\eta v_X Q_{AB}}{\kappa}, \qquad Q_{AB} = N_A N_B, \qquad R_{AB}(N_A, N_B) = \frac{(\beta\eta v_X Q_{AB})^2}{2\kappa}.$$

Derivatives of b^*. Using $Q_{AB} = N_A N_B$,

$$\frac{\partial b^*}{\partial N_A} = \frac{\beta\eta v_X N_B}{\kappa} > 0, \tag{114}$$

$$\frac{\partial b^*}{\partial N_B} = \frac{\beta\eta v_X N_A}{\kappa} > 0, \tag{115}$$

$$\frac{\partial b^*}{\partial \eta} = \frac{\beta v_X Q_{AB}}{\kappa} > 0, \tag{116}$$

$$\frac{\partial b^*}{\partial \kappa} = -\frac{\beta\eta v_X Q_{AB}}{\kappa^2} < 0, \tag{117}$$

$$\frac{\partial b^*}{\partial v_X} = \frac{\beta\eta Q_{AB}}{\kappa} > 0, \tag{118}$$

which are exactly (56)–(60).

Derivatives of $R_{AB}(N_A, N_B)$. Write

$$R_{AB}(N_A, N_B) = \frac{\beta^2\eta^2 v_X^2}{2\kappa} Q_{AB}^2 = \frac{\beta^2\eta^2 v_X^2}{2\kappa} N_A^2 N_B^2.$$

Differentiating with respect to N_A,

$$\frac{\partial R_{AB}(N_A, N_B)}{\partial N_A} = \frac{\beta^2\eta^2 v_X^2}{2\kappa} \cdot 2N_A N_B^2$$
$$= \frac{\beta^2\eta^2 v_X^2 N_A N_B^2}{\kappa} > 0, \tag{119}$$

which is (61).

Differentiating with respect to N_B,

$$\frac{\partial R_{AB}(N_A, N_B)}{\partial N_B} = \frac{\beta^2\eta^2 v_X^2}{2\kappa} \cdot 2N_A^2 N_B$$
$$= \frac{\beta^2\eta^2 v_X^2 N_A^2 N_B}{\kappa} > 0, \tag{120}$$

which is (62).

Differentiating with respect to η,

$$\frac{\partial R_{AB}(N_A, N_B)}{\partial \eta} = \frac{2\beta^2\eta v_X^2 Q_{AB}^2}{2\kappa}$$
$$= \frac{\beta^2\eta v_X^2 Q_{AB}^2}{\kappa} > 0, \tag{121}$$

which is (63).

Differentiating with respect to κ,

$$\frac{\partial R_{AB}(N_A, N_B)}{\partial \kappa} = -\frac{(\beta\eta v_X Q_{AB})^2}{2\kappa^2} < 0, \tag{122}$$

which is (64).

Differentiating with respect to v_X,

$$\frac{\partial R_{AB}(N_A, N_B)}{\partial v_X} = \frac{2\beta^2\eta^2 v_X Q_{AB}^2}{2\kappa}$$
$$= \frac{\beta^2\eta^2 v_X Q_{AB}^2}{\kappa} > 0, \tag{123}$$

which is (65).

This proves Proposition 8. □

D Proofs for Section 7

This appendix provides the algebra for the extensions in Section 7.

D.1 Preliminaries

Recall the baseline objects

$$P_A = \frac{N_A(N_A-1)}{2}, \qquad Q_{AB} = N_A N_B, \qquad \Delta m_A := m_A^1 - m_A^0 > 0, \qquad \Lambda_A := (1+\beta) - \beta\delta(m_A^1 + m_A^0).$$

In the heterogeneous-participant extension, the expected surplus per realized local interaction is

$$\bar{v}_L(\sigma) = v_L^L(1-\sigma)^2 + v_L^H\left[1-(1-\sigma)^2\right] = v_L^L + (2\sigma - \sigma^2)(v_L^H - v_L^L), \tag{124}$$

with

$$\bar{v}_L'(\sigma) = 2(1-\sigma)(v_L^H - v_L^L) > 0. \tag{125}$$

Also, since $\bar{v}_L(\sigma)$ is a convex combination of v_L^L and v_L^H,

$$\bar{v}_L(\sigma) > 0 \qquad \text{for all } \sigma \in [0,1]. \tag{126}$$

The expected surplus per effective cross-regional link is

$$\bar{v}_X(\sigma) = v_X^L(1-\sigma) + v_X^H\sigma = v_X^L + \sigma(v_X^H - v_X^L), \tag{127}$$

with

$$\bar{v}_X'(\sigma) = v_X^H - v_X^L > 0 \tag{128}$$

and

$$\bar{v}_X(\sigma) > 0 \qquad \text{for all } \sigma \in [0,1]. \tag{129}$$

In the online-substitution extension, the no-hub matching rate is

$$m_A^0(o) = m_A^0 + \phi o, \qquad \phi > 0, \tag{130}$$

with domain

$$0 \le o < \bar{o} := \frac{m_A^1 - m_A^0}{\phi}, \tag{131}$$

so that $0 \le m_A^0(o) < m_A^1 \le 1$.

D.2 Proof of Proposition 9

Using (66), the hub-alone welfare gain is

$$\Delta_H(\sigma) = \bar{v}_L(\sigma) P_A \Delta m_A \Lambda_A - F.$$

Differentiating with respect to σ,

$$\frac{\partial \Delta_H(\sigma)}{\partial \sigma} = \bar{v}_L'(\sigma) P_A \Delta m_A \Lambda_A. \tag{132}$$

If $\Lambda_A > 0$, then (125) implies

$$\frac{\partial \Delta_H(\sigma)}{\partial \sigma} > 0 \qquad \text{for } \sigma \in [0, 1),$$

which proves (73).

Next, the optimal pipeline policy is

$$b^*(\sigma) = \frac{\beta\eta\bar{v}_X(\sigma)Q_{AB}}{\kappa}.$$

Differentiating,

$$\frac{\partial b^*(\sigma)}{\partial \sigma} = \frac{\beta\eta Q_{AB}}{\kappa}\bar{v}'_X(\sigma). \tag{133}$$

Since all multiplicative terms are positive and $\bar{v}'_X(\sigma) > 0$, we obtain

$$\frac{\partial b^*(\sigma)}{\partial \sigma} > 0,$$

which proves (74).

For the renewal surplus,

$$R_{AB}(N_A, N_B; \sigma) = \frac{\left(\beta\eta\bar{v}_X(\sigma)Q_{AB}\right)^2}{2\kappa} = \frac{\beta^2\eta^2 Q_{AB}^2}{2\kappa}\,\bar{v}_X(\sigma)^2.$$

Differentiating with respect to σ,

$$\begin{aligned}\frac{\partial R_{AB}(N_A, N_B; \sigma)}{\partial \sigma} &= \frac{\beta^2\eta^2 Q_{AB}^2}{2\kappa} \cdot 2\bar{v}_X(\sigma)\bar{v}'_X(\sigma) \\ &= \frac{\beta^2\eta^2 Q_{AB}^2}{\kappa}\,\bar{v}_X(\sigma)\bar{v}'_X(\sigma).\end{aligned} \tag{134}$$

By (129) and (128), this derivative is strictly positive, proving (75).

Finally, consider the threshold statements. The short-run private-support condition in Section 5 becomes

$$F < \bar{v}_L(\sigma)P_A\Delta m_A.$$

Thus, treating N_A as continuous,

$$N^*_{SR}(\sigma) = \frac{1}{2}\left(1 + \sqrt{1 + \frac{8F}{\bar{v}_L(\sigma)\Delta m_A}}\right). \tag{135}$$

Differentiating,

$$\frac{dN^*_{SR}(\sigma)}{d\sigma} = -\frac{2F\,\bar{v}'_L(\sigma)}{\Delta m_A\,\bar{v}_L(\sigma)^2\sqrt{1 + \frac{8F}{\bar{v}_L(\sigma)\Delta m_A}}} < 0. \tag{136}$$

Likewise, when $\Lambda_A > 0$, the hub-efficiency threshold in Section 4 becomes

$$N_H^*(\sigma) = \frac{1}{2}\left(1 + \sqrt{1 + \frac{8F}{\bar{v}_L(\sigma)\Delta m_A \Lambda_A}}\right). \tag{137}$$

Differentiating,

$$\frac{dN_H^*(\sigma)}{d\sigma} = -\frac{2F\,\bar{v}_L'(\sigma)}{\Delta m_A \Lambda_A\,\bar{v}_L(\sigma)^2\sqrt{1 + \frac{8F}{\bar{v}_L(\sigma)\Delta m_A \Lambda_A}}} < 0. \tag{138}$$

Therefore both thresholds decrease with σ. This completes the proof of Proposition 9. □

D.3 Proof of Proposition 10

In the online-substitution extension,

$$m_A^0(o) = m_A^0 + \phi o, \qquad 0 \le o < \bar{o} = \frac{m_A^1 - m_A^0}{\phi}.$$

The short-run private gain is

$$\Pi_H(o) = v_L P_A\big(m_A^1 - m_A^0(o)\big) - F = v_L P_A\big(m_A^1 - m_A^0 - \phi o\big) - F.$$

Differentiating with respect to o,

$$\frac{d\Pi_H(o)}{do} = -\phi v_L P_A < 0, \tag{139}$$

which proves (81).

Next, the hub-alone welfare gain is

$$\Delta_H(o) = v_L P_A\big[m_A^1 - m_A^0(o)\big]\Big[(1+\beta) - \beta\delta\big(m_A^1 + m_A^0(o)\big)\Big] - F.$$

For brevity, write $x := m_A^0(o)$. Then

$$\Delta_H(o) = v_L P_A(m_A^1 - x)\Big[(1+\beta) - \beta\delta(m_A^1 + x)\Big] - F.$$

Differentiate first with respect to x:

$$\begin{aligned}\frac{d}{dx}\Big[(m_A^1 - x)\big((1+\beta) - \beta\delta(m_A^1 + x)\big)\Big] &= -\Big[(1+\beta) - \beta\delta(m_A^1 + x)\Big] - (m_A^1 - x)\beta\delta \\ &= -(1+\beta) + \beta\delta(m_A^1 + x) - \beta\delta(m_A^1 - x) \\ &= -(1+\beta) + 2\beta\delta x. \end{aligned} \tag{140}$$

Since $dx/do = \phi$, the chain rule gives

$$\begin{aligned}\frac{d\Delta_H(o)}{do} &= \phi v_L P_A\Big[-(1+\beta) + 2\beta\delta m_A^0(o)\Big] \\ &= -\phi v_L P_A\Big[(1+\beta) - 2\beta\delta m_A^0(o)\Big]. \end{aligned} \tag{141}$$

This is exactly (82).

To verify the sign, note that the domain restriction implies $m_A^0(o) < m_A^1$, and the regularity condition (6) implies $\delta m_A^1 < 1$. Hence $\delta m_A^0(o) < 1$ as well. Therefore

$$2\beta\delta m_A^0(o) < 2\beta.$$

Since $\beta \in (0, 1]$, we have $1 + \beta \geq 2\beta$, with equality only when $\beta = 1$. Thus

$$(1+\beta) - 2\beta\delta m_A^0(o) > (1+\beta) - 2\beta = 1 - \beta \geq 0,$$

where the strict inequality follows from $\delta m_A^0(o) < 1$. Hence

$$\frac{d\Delta_H(o)}{do} < 0.$$

Finally, conditional on a hub, welfare with pipeline policy is

$$W(1, b) = W(1, 0) + \beta\eta v_X Q_{AB} b - \frac{\kappa}{2} b^2.$$

The planner therefore solves

$$\max_{b \geq 0} \left\{ W(1, 0) + \beta\eta v_X Q_{AB} b - \frac{\kappa}{2} b^2 \right\}.$$

The first-order condition is

$$\beta\eta v_X Q_{AB} - \kappa b = 0,$$

so

$$b^*(o) = \frac{\beta\eta v_X Q_{AB}}{\kappa} = b^*.$$

Thus the optimal pipeline policy is unchanged by online substitution in this reduced form, which proves (83). This completes the proof of Proposition 10. □